

Abstract

Robots that handle fabric — for garment sorting, textile recycling, or assistive dressing — need a sense of touch, because two garments that look similar can feel entirely different. A recently published framework pairs a vision-based tactile sensor with a multimodal large language model so that a robot can press two fabrics and state which one is softer, thicker, more elastic, or more textured. The public release of that framework contains the data, but the training code, the model weights and the answer-extraction procedure are not recorded in the released artifacts, so any laboratory wishing to build on the work must first rebuild it. This project asks three questions. Can the published system be reproduced independently from its released artifacts alone? What does its evaluation protocol measure, and how do the numbers change under alternative protocols? And can the system be deployed on a real robot arm when no force sensor is available in the deployment loop?

The reproduction was rebuilt from the ground up: a single prompt-generation module verified byte-for-byte against all 8,734 released prompts, a calibrated free-text answer parser with a measured noise floor of 1.94 percentage points, and a tie-free forced-choice scorer computed in full numerical precision. A seven-billion-parameter vision–language model fine-tuned with quantised low-rank adapters reached 0.9850 forced-choice accuracy on the official test protocol, whose fabric-overlap structure admits a level look-up upper bound of 1.00; every accuracy in this report is therefore presented next to the look-up ceiling of its own pair of training and test sets. Three extensions follow. First, a fabric-disjoint evaluation split — domain-matched by construction and carrying a 0.50 ceiling — measures generalisation to unseen fabrics: the same configuration scores 0.8650, and the gap to a size-matched control (+0.1200) concentrates almost entirely in a single property. Second, the task is made finer-grained: 63,300 adjacent-level comparison pairs are enumerated from the released annotations without any new labelling, and adding 4,000 of them to training raises adjacent-level accuracy from 0.8387 to 0.9750. Third, a load-estimation network trained on a separately collected dataset of 12,178 force-calibrated presses replaces the force sensor at deployment (mean absolute error 0.3256 N on held-out fabrics), and the complete pipeline — press, estimate force, select frames, query the model, rank four garments and recommend one — runs end-to-end on a robot arm with vision-based tactile sensing. The project shows that the published task is reproducible from its artifacts, quantifies how strongly the reported numbers depend on the evaluation protocol, and demonstrates a deployment path that requires no force sensor at run time.

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List of Symbols

Roman symbols

F	normal contact force (N), measured by the ATI Nano17 reference sensor
$\hat{F}$	normal contact force estimated from a GelSight image (N)
F_{peak}	peak force of a press (N)
$g_p(f)$	ordinal level of fabric f for property p , $g_p(f) \in \{0, 1, 2\}$
I	composite (concatenated) input image
y	comparison label: the fabric with the higher property level
ℓ_k	log-likelihood of candidate continuation k under the model
m	forced-choice margin, $m = \ell_1 - \ell_2 $
c	look-up coverage of a (training set, test set) pair
U	look-up-table accuracy ceiling of an evaluation cell
W_0	frozen pre-trained weight matrix
B, A	low-rank adapter factors, $\Delta W = \frac{\alpha}{r} BA$
r, α	adapter rank and scaling factor
n	sample size
d	Cohen's d effect size
$T(n)$	wall-clock evaluation time for n samples (s)

Greek symbols

τ	camera–force synchronisation delay (ms)
σ	standard deviation
ρ	Spearman rank correlation
ω^2	analysis-of-variance effect size
θ, ϕ	parameters of the language model and the force estimator

Acronyms

AUC	area under the receiver-operating-characteristic curve
bf16	bfloat16 floating-point format
fp32	IEEE-754 single-precision floating-point format
ICC	intraclass correlation coefficient
LoRA	low-rank adaptation
MAE	mean absolute error
MLLM	multimodal large language model
OLS	ordinary least squares
QLoRA	quantised low-rank adaptation
RGB	red–green–blue (visible-light image)
RMSE	root-mean-square error
ULP	unit in the last place (floating-point spacing)
VLM	vision–language model

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Chapter 1: Introduction

1.1 Background

Fabric is one of the most common materials a robot will ever touch, and one of the hardest to perceive. Garment sorting for recycling, laundry automation, textile quality control and assistive dressing all require judgements — *which of these two garments is softer, thicker, more elastic, rougher?* — that colour images alone often cannot support: two fabrics that photograph almost identically can differ strongly in weave, pile and compliance [2–4]. Deformable-object manipulation is repeatedly identified as one of the open challenges of robotic manipulation [5], and touch is the modality that exposes exactly the mechanical properties vision misses [6, 7]. Vision-based tactile sensors such as GelSight [8, 9] address the sensing side by imaging the deformation of an illuminated elastomer pressed against a surface, turning touch into high-resolution images that standard computer-vision models can process [10, 11]. In parallel, multimodal large language models (MLLMs) have made it possible to connect such images to natural language, so that a robot can not only sense a material but also answer questions about it [12–15].

MLLM-Fabric [1] combines these two developments. It pairs RGB photographs of fabrics with GelSight images captured at increasing contact forces, and fine-tunes a vision–language model (VLM) to compare two fabrics along four properties — elasticity, softness, texture and thickness — each annotated on a three-level ordinal scale. The public release comprises roughly 3.6 GB of images and conversation-formatted training and test files covering 200 fabrics, 3,937 training samples and 400 test samples.

1.2 Problem Statement

Three practical obstacles stand between the published work and a laboratory that wants to build on it.

First, the released artifacts contain the data but do not record the training code, the model weights, or the procedure used to extract a discrete answer from the model’s free-text output. The reported accuracy therefore cannot be recomputed without rebuilding the entire stack, and a rebuilt stack must prove that its inputs match the originals exactly, because the model is trained on the exact bytes of the prompts — including three formatting irregularities that a well-meaning reimplementer would silently “correct”.

Second, the evaluation protocol has measurable structural properties that determine what a reported accuracy means. The training and test sets share 186 of 200 fabrics, and the pairs published for testing all compare fabrics whose property levels differ by two grades. As Chapter 3 quantifies, a level look-up table built from the training pairs alone — without looking at a single pixel — answers all 400 official test items correctly. A reported accuracy on this protocol must therefore be read against a memorisation ceiling of 1.00, and the protocol carries no information about the harder adjacent-level comparisons. This is not a defect peculiar to one dataset: shortcut opportunities and annotation artefacts of exactly this kind are documented across machine-learning benchmarks [16–19], and for large language models the corresponding failure is benchmark contamination [20–22]. Measuring these properties, and constructing complementary protocols with different ceilings, is a prerequisite

for interpreting any number this report produces.

Third, the data-collection rig assumes a calibrated force/torque sensor beneath the fabric, but a deployed robot pressing hanging garments has no such sensor in the loop; the force values that the prompt format requires must come from somewhere else. Without a sensor-free deployment path the system remains confined to its data-collection rig.

1.3 Proposed Solution

This project reproduces MLLM-Fabric independently, from the released artifacts alone, and extends it in three directions. The problem is modelled throughout as pairwise ordinal comparison under two complementary readouts — free-text generation scored by a calibrated parser, and forced choice scored by candidate log-likelihoods in full precision — and every accuracy is reported next to the look-up ceiling of its own (training set, test set) pair.

The reproduction is full-stack: a single prompt-generation module verified byte-for-byte against all 8,734 released prompts; an image-composition pipeline shown to be equivalent to the released layout; QLoRA fine-tuning [23, 24] of Qwen2.5-VL-7B-Instruct [14]; a calibrated free-text answer parser with a measured noise floor; and a tie-free forced-choice scorer.

The three extensions are:

1. **Evaluation protocol (T1).** A fabric-disjoint evaluation split inside the original 200 fabrics, constructed to satisfy two pre-registered criteria (zero look-up coverage and no detectable acquisition-domain signal), measures generalisation to unseen fabrics; a size-matched control isolates the contribution of having seen the test fabrics during training.
2. **Task granularity (T3).** The released pairs all differ by two grades; 63,300 adjacent-level (one-grade) pairs are enumerated from the released annotations without any new labelling, enabling finer-grained training and testing, a controlled ablation of rationale leakage in the supervision signal, a cross-domain test on fabrics from a different acquisition batch, and direct perturbation probes of whether the model reads the force values in its prompt.
3. **Real-robot deployment (T-D0).** A load-estimation network trained on a separately collected dataset (200 fabrics, 12,178 calibrated presses) supplies the force values at deployment, replacing the force sensor entirely; a teach-and-replay pressing routine, gated frame selection and an unchanged cloud evaluator complete an end-to-end pipeline demonstrated on four hanging garments with a Franka arm [25].

The headline results are: forced-choice accuracy 0.9850 on the official protocol (ceiling 1.00); 0.8650 on the fabric-disjoint protocol (ceiling 0.50), with the gap to the size-matched control (+0.1200) concentrated almost entirely in one property; adjacent-level accuracy raised from 0.8387 to 0.9750 (cell ceilings 0.50 and 0.9988 respectively) by adding 4,000 enumerated pairs; a deployed force estimator with mean absolute error 0.3256 N; and an end-to-end robot demonstration that ranks four garments and issues a natural-language recommendation with per-property reliability warnings.

1.4 Report Outline

The rest of this report is organised as follows. Chapter 2 reviews tactile sensing, visuo–tactile learning, tactile-language models and evaluation methodology. Chapter 3 formalises the task, defines the look-up ceiling, and states the research questions, assumptions and objectives. Chapter 4 describes the reproduction stack, the evaluation instruments, the construction of the extension protocols and the deployment pipeline. Chapter 5 presents and analyses all experimental results, including the measured outcome of the force-sensor commissioning check. Chapter 6 evaluates the project against its objectives and discusses impact and limitations, and Chapter 7 concludes with recommendations for further work. Appendices A–C record protocol details, the reproducibility ledger and supplementary results.

Chapter 2: Literature Review

This chapter reviews the four bodies of work the project draws on: tactile sensing hardware (§2.1), visuo–tactile machine learning and fabric perception (§2.2), vision–language models and tactile-language systems (§2.3), and the evaluation-methodology literature that motivates the protocol-measurement stance taken throughout this report (§2.5).

2.1 Tactile sensing for robotic perception

Robotic tactile sensing has a long history of transduction principles — resistive, capacitive, piezo-electric, optical — surveyed comprehensively by Dahiya *et al.* [6] and, for dexterous hands specifically, by Kappassov *et al.* [7]. The line of work this project builds on is *camera-based* tactile sensing, which trades electrical taxel arrays for an internal camera observing a deformable membrane. Johnson and Adelson’s retrographic sensing [8] established the principle; the GelSight family developed it into a practical sensor whose images encode fine surface geometry and, through marker motion and membrane deformation, contact force [9]. Subsequent designs pursued miniaturisation and cost (DIGIT [26]) or soft biomimetic morphologies (TacTip [27]). Reviews of the resulting design space [10, 11] emphasise a point that matters for this project: the image produced by such a sensor is not an invariant physical measurement but a function of its illumination, elastomer batch, optics and image pipeline. Chapter 5 shows this concretely — a linear probe separates two acquisition batches of the same benchmark with AUC 1.0000 even after photometric correction.

A GelSight image also carries force information. Prior work has estimated contact force from marker displacement fields and membrane deformation [9], and learned grasp-relevant quantities directly from tactile images [28]. This project follows the learned route: a ResNet-18 [29] regressor maps a single photometrically reconstructed frame to absolute normal force, trained against a reference force/torque sensor and deployed where that sensor is absent.

2.2 Visuo–tactile learning and fabric perception

Paired visual–tactile data enable representations that connect how surfaces look with how they feel. Yuan *et al.* associated visual and tactile observations of clothing [30] and demonstrated active clothing material perception with a GelSight-equipped robot [2]; Luo *et al.* showed that sharing features between vision and touch improves cloth texture recognition [31]. Larger multimodal corpora followed: Touch and Go collected human visuo–tactile recordings in the wild [32], ObjectFolder 2.0 provided implicit multisensory object representations with real-object counterparts [33], and visuo–tactile self-supervision has been used to locate and follow garment features [34]. Two observations from this literature shape the present work: (i) fabric and clothing are among the most-studied targets for visuo–tactile perception because their mechanical properties are weakly visible; and (ii) transfer across sensors and acquisition set-ups is a recurring difficulty — learned tactile representations inherit the device that produced them.

2.3 Vision–language models and tactile-language systems

Contrastive image–text pretraining [12] and transformer vision backbones [35] led to instruction-following vision–language models such as LLaVA [13] and GPT-4V [36]. The Qwen-VL series is of particular

relevance: Qwen2-VL introduced native dynamic resolution, mapping images of arbitrary size to a bounded number of visual tokens [37], and Qwen2.5-VL — the base model used here — retained this mechanism while improving recognition and document understanding [14]. Dynamic resolution has a direct experimental consequence exploited in Chapter 4: beyond the processor’s pixel budget, a larger stored image adds no model-visible resolution, which permits a $6.6\times$ reduction in upload volume with pixel-level equivalence.

Bridging language models and robotics, PaLM-E injected multimodal embeddings into a language model for embodied tasks [38] and RT-2 transferred web-scale vision–language knowledge to robotic control [39]. Touch entered this picture recently: Octopi couples a tactile sensor with a language model for object-property reasoning [15]; the Touch–Vision–Language dataset aligns tactile images with captions for multimodal training [40]; and unified tactile encoders spanning multiple sensors have been proposed [41]. MLLM-Fabric [1] — the object of this reproduction — applies this paradigm to fabric: RGB and multi-force GelSight images are composed into a single prompt, and a fine-tuned VLM (distilled from a larger teacher [42]) answers pairwise property comparisons in natural language. Its released artifacts are the sole input to the present reproduction; Table 1 positions this project among the works above.

Table 1 Position of this project relative to representative prior work.

Work	Modalities	Relation to this project
GelSight [9]	tactile image	sensing principle for both data pipelines
Active clothing perception [2]	tactile + robot	fabric-property precedent, fixed classifier head
Octopi [15]	tactile + language	property reasoning via an LLM, different sensor and task
TVL [40]	touch + vision + text	alignment corpus; no ordinal comparison task
MLLM-Fabric [1]	RGB + tactile + force + text	the reproduced system
This project	same released inputs; separate force pipeline at deployment	independent reproduction; ceiling-aware evaluation; adjacent-level construction; sensor-free deployment

2.4 Parameter-efficient adaptation and numerical precision

Full fine-tuning of a 7-billion-parameter model is memory-prohibitive on a single GPU and risks overwriting pretrained visual features. Low-rank adaptation (LoRA) freezes the pretrained matrix W_0 and learns an additive low-rank update [23]; QLoRA further quantises the frozen base to 4 bits while training the adapters in higher precision [24]. Training uses the AdamW optimiser [43]. Reduced-precision arithmetic, standard for training [44], has a sharp edge at evaluation time: bfloat16 keeps float32’s exponent range but only an 8-bit significand [45], so its representable values near a log-likelihood sum of magnitude 16–32 are 0.125 apart. Chapter 4 shows why this makes fp32 mandatory for forced-choice scoring — a lesson that, to the author’s knowledge, is not spelled out in the tactile-language literature.

2.5 Evaluation methodology: what does an accuracy measure?

A model score is only interpretable relative to what the evaluation data could have leaked from training. This concern predates language models: Torralba and Efros documented dataset bias in vision benchmarks [16]; Recht *et al.* showed that new, protocol-faithful ImageNet test sets shift accuracies substantially [46]; annotation artefacts allowed hypothesis-only models to pass NLI benchmarks [17]; and spurious cues gave BERT near-human argument comprehension that vanished when the cues were removed [18]. Geirhos *et al.* unify these observations as *shortcut learning* [19]. For large language models, train–test leakage becomes benchmark contamination [20, 21], and statistical tests now exist to detect it from model behaviour alone [22]. Holistic evaluation frameworks recommend multi-metric, scenario-explicit reporting [47].

The methodological stance of this report follows directly from that literature, in three commitments. First, the shortcut structure of the evaluated benchmark is *measured* — as a look-up ceiling per (training set, test set) pair — rather than assumed absent. Second, complementary protocols with different ceilings (fabric-disjoint, adjacent-level, cross-domain) are constructed so that different generalisation questions receive different measurements. Third, the answer-extraction step is treated as part of the instrument: free-text parsing is calibrated with a measured noise floor, and a parser-free forced-choice readout is reported alongside it. Effect sizes accompany hypothesis tests throughout, following standard reporting recommendations [48].

2.6 Summary of the gap

The literature provides mature tactile sensors, strong pretrained VLMs, and clear warnings about shortcut-prone evaluation — but no published account of independently rebuilding a tactile-language benchmark from its artifacts, quantifying its protocol ceilings, and carrying the result onto a robot without the data-collection rig’s force sensor. That combination is the contribution of this project.

Chapter 3: Problem Formulation

3.1 Task definition

Let $\mathcal{P} = \{\text{Elasticity, Softness, Texture, Thickness}\}$ be the property set. Every fabric f in the released annotation table carries an ordinal level $g_p(f) \in \{0, 1, 2\}$ for each property $p \in \mathcal{P}$. One evaluation item presents two fabrics (f_1, f_2) with $g_p(f_1) \neq g_p(f_2)$, a composite image I , and a natural-language question naming p ; the target is

$$y = \arg \max_{i \in \{1, 2\}} g_p(f_i), \quad (1)$$

i.e. the fabric with the higher level. The composite image has two rows (one per fabric) and five columns: an RGB view followed by four GelSight frames captured at increasing contact force, whose numeric values also appear in the prompt text. The released protocol uses only pairs with $|g_p(f_1) - g_p(f_2)| = 2$; the adjacent-level extension constructed in this project uses $|g_p(f_1) - g_p(f_2)| = 1$. Where a test set contains both orders of each unordered pair, a symmetry decomposition applies (Appendix A): a pair can be ranked correctly in both orders (2/2), reversed consistently (0/2), or answered by row position alone (1/2), so 0.5 row accuracy does not distinguish “no information” from “always pick the same position”.

3.2 Two readouts

The model under test is generative, so a discrete answer must be obtained in one of two ways.

Free text. The model generates an explanation; a parser extracts a verdict. Three quantities are always reported together: with n_{parse} parsed and n_{corr} correct responses out of n ,

$$\text{acc}_{\text{all}} = \frac{n_{\text{corr}}}{n}, \quad \text{acc}_{\text{parsed}} = \frac{n_{\text{corr}}}{n_{\text{parse}}}, \quad r_{\text{parse}} = \frac{n_{\text{parse}}}{n}. \quad (2)$$

Reporting any one of these alone hides either abstentions or parser failures.

Forced choice. The log-likelihood of each candidate continuation $k \in \{1, 2\}$ is accumulated over its tokens,

$$\ell_k = \sum_{t=1}^{T_k} \log P_{\theta}(a_{k,t} \mid a_{k,<t}, I, q), \quad \hat{y} = \arg \max_k \ell_k, \quad m = |\ell_1 - \ell_2|, \quad (3)$$

with margin m retained per item. This readout needs no parser, but it is numerically fragile: in bfloat16 the unit in the last place for values in $[16, 32)$ is

$$\text{ULP}_{\text{bf16}}(x) = 2^{\lfloor \log_2 x \rfloor - 7} = 0.125 \quad \text{for } x \in [16, 32), \quad (4)$$

while genuine candidate differences are often 0.01–0.2. Scoring in bf16 therefore collapses real differences into exact ties, which an $\arg \max$ silently resolves to the first index — turning label order into fake accuracy. All forced-choice scores in this report are computed in fp32, and exact ties are recorded rather than resolved.

3.3 The look-up ceiling

The central protocol measurement is a bound computed from the training data alone. Decisive training pairs reveal fabric levels (a $|\Delta| = 2$ pair pins both fabrics’ levels for that property); accumulating these yields a partial table $\hat{g}_p(\cdot)$. For a (training set, test set) pair, let coverage $c \in [0, 1]$ be the fraction of test items whose answer is determined by the table. Guessing the remainder at random gives the *look-up ceiling*

$$U = c \cdot 1 + (1 - c) \cdot \frac{1}{2} = \frac{1 + c}{2}. \quad (5)$$

U is not an upper bound on a visual model — a model that reads the images can exceed it. It states how far a model could get *without* reading them. Three exact facts anchor the report: for the official pair of sets, the table answers 400/400 test items ($c = 1$, $U = 1.00$); for the constructed fabric-disjoint split, $c = 0$ and $U = 0.50$; and official training pairs cover 0 of the 800 adjacent-level cells, so the adjacent test also has $U = 0.50$ under official training. Every accuracy in Chapters 5 appears beside the U of its own cell.

3.4 Research questions

RQ1 — Reproduction.

What is required to reproduce the released pairwise ranking task from its artifacts alone, and does the reproduction reach the published performance regime on the official protocol?

RQ2 — Protocol.

How does measured performance change when the evaluation protocol changes — fabrics unseen but domain-matched, seen with matched training volume, or drawn from a different acquisition batch — and how much of the official-protocol score is attributable to fabric identity rather than tactile comparison?

RQ3 — Granularity.

Can finer (adjacent-level) comparisons be constructed without new annotation, does training on them help, and what do controlled ablations (rationale leakage, side-index reversal, force-text perturbation) reveal about what the model actually uses?

RQ4 — Deployment.

Can the system run on a real robot with no force sensor in the loop, with every number in the prompt traceable to a measurement or an asserted physical fact?

3.5 Assumptions and scope

1. The released property levels are the target definition; no fabric is relabelled.
2. The two data pipelines are strictly separated: no self-collected image enters VLM training or evaluation, and no released image enters force-estimator training. They meet only at robot inference time.
3. Self-collected force labels are trusted up to 22 N (an empirically established limit, Chapter 4); the deployment force ladder therefore stays below 20 N.
4. All VLM runs use a single seed (20260801); observed differences are reported as measurements without variance estimates and are never used to rank close alternatives. The force estimator uses three seeds where an uncertainty is quoted.

5. The four-garment robot demonstration has no independent property ground truth; it is an integration and traceability test, not an accuracy experiment.
6. The Nano17 V2 commissioning check was designed, tooled, and executed; its measured outcome is reported in Section 5.6.

3.6 Objectives

Table 2 Objectives and pre-stated success criteria.

ID	Objective	Success criterion
O1	Rebuild the released task exactly	all 8,734 released prompts byte-identical from one generator; training and evaluation run under pinned versions and a fixed seed
O2	Make the evaluation instrument explicit	every accuracy reported with n and the cell's look-up ceiling; free-text metrics reported as the triple of Eq. (2); forced choice tie-free in fp32
O3	Measure unseen-fabric generalisation	a split with $c = 0$ and an acquisition-domain probe at chance, plus a size-matched control
O4	Extend task granularity	adjacent-level pairs constructed with zero new annotation; effect of adjacent training measured
O5	Deploy a sensor-free force interface	load-estimation $\text{MAE} \leq 0.5 \text{ N}$ overall and $\leq 1.0 \text{ N}$ on 15–22 N loading frames, verified on the deployment inference path
O6	Run an end-to-end traceable robot demonstration	press-to-recommendation chain executes with hash-linked provenance; no unsupported accuracy claim

Chapter 6 returns to Table 2 and assesses each criterion against the evidence of Chapter 5.

Chapter 4: Methodology

4.1 Architecture: two pipelines that meet only at deployment

The single most important design constraint is the strict separation of two data pipelines (Figure 1). Pipeline A is the *self-collected* force dataset: 200 fabrics pressed by a GelSight Mini mounted on a Franka arm [25], with an ATI Nano17 force/torque sensor under the fabric providing ground-truth normal force. It trains one thing only — the load-estimation network. Pipeline B is the *released* MLLM-Fabric data [1]: every VLM training run and every VLM evaluation in this report uses released images and labels exclusively. The pipelines meet for the first time inside the robot demonstration, where a Pipeline-A-trained estimator converts a freshly captured GelSight frame into the force value written into a Pipeline-B-formatted prompt.

This separation closes off two tempting shortcuts. Self-collected fabrics have no property labels, so pseudo-labelling them for VLM training would manufacture supervision; and the two force scales overlap only at the low end (released prompt ladder medians 0, 11.61, 35.12 and 51.98 N against a 22 N trustworthy self-collected ceiling), so pretending the scales are interchangeable would corrupt both.

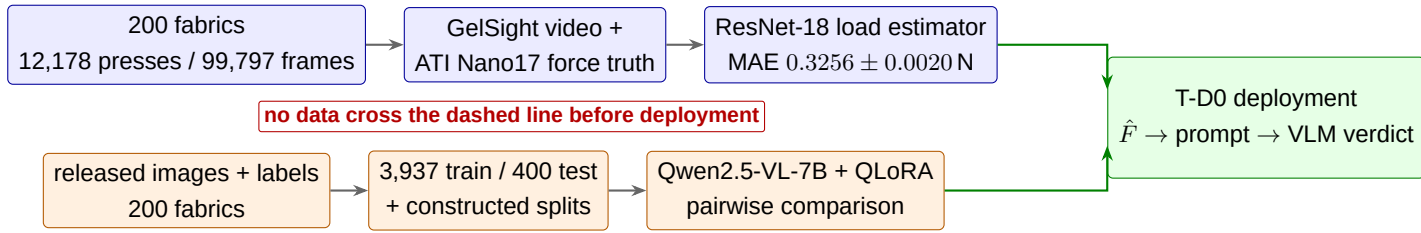


Figure 1 The two isolated pipelines. Pipeline A (top, blue) trains only the load estimator; Pipeline B (bottom, orange) trains and evaluates only the VLM. They meet for the first time at robot inference (green), where the estimator supplies the force text of a prompt whose format is byte-compatible with Pipeline B.

4.2 Reproducing the released interface

4.2.1 Prompt format contract

The released conversation files use a fixed template with three formatting irregularities: two lines ending in a trailing space, an isolated 0 preceding one force slot that appears in training prompts only, and the Texture-specific string `features(such` missing its space. Because the model is trained on these exact bytes — and whitespace changes tokenisation — the reproduction treats them as part of the interface. A single module generates every prompt in the project; a verification gate rebuilds all 8,734 released prompts and compares them byte-for-byte, and applies the same format contract to all 34,883 constructed samples. Nothing in the project concatenates prompt strings anywhere else.

4.2.2 Image pipeline

Each released item is a 5120×2048 composite: 2 rows $\times$ 5 columns of 1024^2 cells (column 0 RGB, columns 1–4 GelSight at increasing force). Qwen2.5-VL’s dynamic-resolution processor [14, 37] resizes any input to at most `max_pixels` while preserving aspect ratio on a 28-pixel grid, so the model-visible cell resolution is set by the token budget, not the stored file. At the chosen budget

of `max_pixels = 1,176,000` (1,464 visual tokens), a 2560×1024 source and the original 5120×2048 both map to 341 pixels per cell — verified by computing the processor’s resize arithmetic explicitly — so all images are shipped at 2560×1024 (LANCZOS downsample, JPEG quality 90), reducing the cloud upload from 11.76 GB to 1.77 GB with an end-to-end reconstruction fidelity of 40.2 dB PSNR. Figure 2 shows a deployment-time composite produced by the same code path.

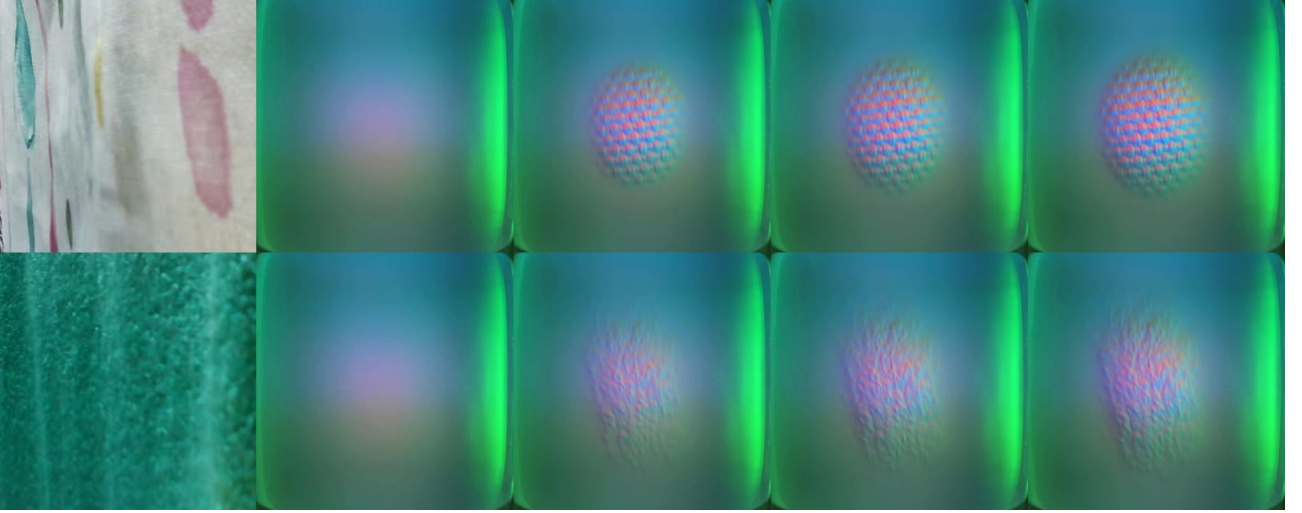


Figure 2 An actual composite input from the robot demonstration (fabrics F1 and F2, forward order). Each row is one fabric: an RGB view, then GelSight frames selected at the force ladder 0/13.5/16.8/19.9 N. The reverse-order control swaps both the image rows and the prompt roles; swapping only the text would produce a semantically invalid sample.

4.2.3 Model adaptation

The base model is Qwen2.5-VL-7B-Instruct [14] with its vision tower frozen. Adaptation uses QLoRA [24]: the base is loaded in 4-bit quantisation and LoRA adapters [23] of rank $r = 32$ are trained,

$$W' = W_0 + \frac{\alpha}{r} BA, \quad B \in \mathbb{R}^{k \times r}, \quad A \in \mathbb{R}^{r \times d}, \quad r \ll \min(k, d), \quad (6)$$

giving 80.7 M trainable parameters. All runs share one configuration — 3 epochs, batch size 1 with gradient accumulation 8, learning rate 10^{-4} under AdamW [43], seed 20260801, `max_pixels = 1,176,000` — and differ *only* in training data (Table 3). Software versions are pinned (transformers 4.49.0, PEFT 0.14.0, PyTorch 2.4.0+cu124); Appendix B lists the full ledger.

Table 3 Training runs. All hyperparameters identical; only the training data differ.

Run	n_{train}	Training data
zero-shot	0	base model, no adaptation (control)
baseline	3,937	all released training records
heldout	2,384	released records excluding the 40 held-out test fabrics
size-matched	2,384	released records, test-fabric coverage retained (volume control)
combined	7,937	released records + 4,000 constructed adjacent-level records
adj-only	4,000	constructed adjacent-level records only
leak-leaky	1,670	records whose rationale restates the verdict
leak-clean	1,670	records whose rationale does not

4.3 Evaluation instruments

4.3.1 Calibrated free-text parser

The released artifacts do not record an answer-extraction method, and none of the 3,937 training rationales contains a structured answer line, so a parser had to be built and — crucially — calibrated. Calibration against all released rationales gives parse rate 0.9050 and parsed-accuracy 0.9806; a 100-item manual adjudication agrees at 0.9785. The resulting noise floor of 1.94 percentage points accompanies every free-text accuracy. The first parser version, before calibration, carried a 6.36-point systematic bias — direct evidence that extraction is part of the measuring instrument, not a formality.

4.3.2 Tie-free forced choice

Forced choice follows Eq. (3) with scoring in fp32; Eq. (4) is the reason. In an early bf16 run the two candidate scores collapsed into exact ties on most items and the arg max returned the first candidate, producing a plausible-looking but fake accuracy of 0.75. After the fp32 fix, all 7,600 scored T1 items contain zero exact ties. Margins are archived per item and reused by the perturbation probes below.

4.3.3 Per-cell ceilings

For every (training set, test set) pair used anywhere in the project, the coverage c and ceiling U of Eq. (5) are computed and stored next to the run outputs, so no accuracy can be quoted without its bound. Cells with different training data have different ceilings even on the same test set — the two rationale-ablation runs differ by 1.12 percentage points in ceiling, which matters when comparing their accuracies.

4.4 T1: the evaluation-protocol extension

Fabric-disjoint split. 200 released fabrics are split 160/40 such that (i) no training pair involves a test fabric, giving $c = 0$ and $U = 0.50$ by construction, and (ii) a learned probe cannot tell the two sides apart. Criterion (ii) is tested with a fabric-held-out ResNet-18 domain classifier: AUC 0.5079 ± 0.0533 (chance), while the same probe code achieves 1.0000 on a known acquisition change — a positive control showing the probe is capable of detecting real domain differences. A naive split that maximised pair capacity failed criterion (ii) (probe 0.5931); appearance balance had to be an explicit constraint. The resulting test sets are *heldout gap-2* (400 items) and *heldout adjacent* (800 ordered items).

Size-matched control. Removing the 40 test fabrics also shrinks training to 2,384 records, so fabric novelty and training volume would otherwise be confounded. The size-matched run keeps exactly 2,384 records *with* test-fabric coverage: comparing heldout to size-matched isolates fabric exposure; comparing size-matched to baseline isolates volume.

4.5 T3: the task-granularity extension

Adjacent-level construction. The released annotations already determine every adjacent comparison: enumerating ordered pairs with $|\Delta g| = 1$ yields 63,300 candidates across the four properties, with zero new annotation. 4,000 are sampled for training (the *combined* and *adj-only* runs) and balanced two-order test sets of 800 items are built for both the shared-fabric and heldout regimes.

Rationale-leakage ablation. 42.42% of released training rationales (1,670 of 3,937, in 1,662 surface variants) restate the final verdict inside the explanation. Whether such restatement helps or harms is treated as an experiment: two runs train on the 1,670 restating records and on 1,670 non-restating records respectively. Because the two subsets cover different fabric–property cells, their ceilings differ (0.9788 vs 0.9900) and are always co-reported.

Cross-domain test. Released fabrics 201–220 come from a visibly different acquisition batch: a fabric-held-out domain probe separates them from fabrics 1–200 with AUC 1.0000, and three photometric corrections reduce the largest univariate effect size from $|d| = 46.97$ to 1.89 without moving the AUC. Cross-domain cells are therefore labelled “capability + domain shift” and never used as clean capability controls.

Perturbation probes. Two further tests reuse trained adapters and move exactly one thing. The *side-swap* test regenerates the official 400 items with press-side indices exchanged, images and answers untouched. The *force-text* probes rewrite only the four force strings: variant C leaves them untouched (and is verified MD5-identical to the released test file — the transparency control), variant A substitutes the deployment ladder 10.20/13.50/16.80/19.90 N, and variant B differs from A in slot 1 only (0.00 N). The sensitivity verdict is judged by a pre-registered rule (prediction changes, sign-test on margin movement, and relative margin effect), together with an arithmetic power check: if the largest margin perturbation is smaller than the smallest identity margin, zero flips are guaranteed and must be reported as such, not as proof of insensitivity.

4.6 T-D0: the deployment extension

4.6.1 Self-collected dataset and its calibrations

Pipeline A contains 401 sessions, 12,239 presses and 99,797 usable frames over 200 fabrics after quality filtering. Three calibrations underpin its use. *Synchronisation*: metadata claimed a 20 ms camera–force delay; cross-correlating force with image change over 120 presses measured $\tau = 40$ ms ($r = 0.9933$, $\sigma = 4.16$ ms across sessions), and applying the correction moved the peak frame in 38% of presses. *Trustworthy force range*: although the Nano17 F_z rating is ± 17 N, three independent checks (contact-area consistency, residual shape, evidence density) support using loading data to 22 N; evidence density collapses above it (Figure 15 in Chapter 5). *Photometric reconstruction*: three pre-stated criteria pass, but the pixel-noise criterion passes marginally (median coefficient of variation 14.87% against a 15% threshold, four of eight groups passing), so single-frame reconstructed magnitudes are never used as absolute labels.

4.6.2 Load estimator

A ResNet-18 [29] (ImageNet-initialised [49]) maps one 224×224 photometrically reconstructed frame to absolute force $\hat{F} = h_\phi(I)$, with an auxiliary four-level ladder head. The split is by fabric (140/30/30), never by frame. Checkpoint selection proved decisive: selecting by validation force MAE instead of the auxiliary ladder accuracy reduced MAE from 0.4328 N to 0.3256 ± 0.0020 N over three seeds (-22.3% , Cohen’s $d = 2.35$), and a controlled decomposition attributes 87% of that improvement (0.0817 N) to the selection criterion alone — the force-optimal epochs (13, 14, 14) lie far from the ladder-optimal ones (5, 10, 5).

4.6.3 End-effector and gripper design

Grasping and pressing hanging garments imposes requirements that the stock parallel-jaw fingers do not meet, so a custom end-effector was designed and 3D-printed (Figure 3). It extends the Franka Hand with two asymmetric fingers and integrates all three sensing and interaction functions of the demonstration in one tool — no tool change occurs between locating, pressing and grasping.

Three design decisions carry the load:

1. **Slender, tapered fingers for single-layer access.** A garment on a hanger is a multi-layer, closely spaced object: the finger that grasps it must pass *through* the garment's opening (or between adjacent garments) and isolate a single layer of fabric without dragging its neighbours. Both fingers therefore taper towards thin, blade-like profiles that can be inserted through the middle of a hanging garment before the jaws close, which the stock blunt fingertips cannot do without bunching the fabric.
2. **Compliant printed tips as sacrificial front ends.** The frontmost segment of each finger is a separately printed, ribbed compliant tip. During insertion the fingers inevitably risk striking rigid objects — the hanger, the rail, the clip — and the GelSight Mini's elastomer pad and internal optics are the most fragile and expensive parts of the tool. The compliant tips deflect on unintended contact and are the first (and cheapest) part to yield or break, while the sensor sits recessed behind them so that rigid obstacles meet the soft tip before they can meet the gel. The same compliance also protects the garment from snagging.
3. **An opposing backing finger for defined presses.** A freely hanging garment yields under load, which would make the contact condition of a tactile press ill-defined and unlike the flat-substrate presses of the training data. The broad paddle face of the opposing finger acts as a backing surface: closing the jaws pinches the fabric between the GelSight face and the paddle, so the press happens against a defined, repeatable substrate. The GelSight Mini is mounted mid-finger with its sensing face inward for exactly this reason, and the wrist camera is integrated on the same printed mount for visual servoing to the garment.

The tips and finger bodies are printed parts and are treated as consumables; the design files sit in the repository alongside the deployment code. No quantitative impact-survival claim is made for the tips — their role in this project is preventive.



Figure 3 The custom end-effector. Two 3D-printed asymmetric fingers extend the Franka Hand: the right finger carries the GelSight Mini mid-finger with its sensing face inward, recessed behind a ribbed compliant tip; the left finger is a broad backing paddle with a matching compliant tip. The wrist camera is integrated on the same mount (bottom). The tapered profiles let the fingers pass through a hanging garment to pinch a single fabric layer between the sensor face and the paddle, giving presses a defined substrate; the compliant tips are the sacrificial first contact if the fingers strike the hanger or rail, protecting the sensor’s elastomer and optics.

4.6.4 Robot pipeline

Figure 4 shows the deployment chain. A press trajectory is taught once under zero position stiffness (an operator physically guides the arm) and validated against four recorded-trajectory checks before being accepted; replay then executes the stored trajectory — with a hard stop able to halt the arm within one 20 ms control cycle — while GelSight frames are captured (MJPG 3280×2464 , area-downsampled to 640×480 , matching the training recordings’ delivered frame rate to 0.01%). Every frame receives a force estimate; a slot selector picks one pre-contact frame and three loading frames nearest the ladder targets, subject to six gates (monotonicity, target tolerance, spacing, loading-phase membership, the 22 N cap, and duration; Appendix A).

Slot 1 is asserted as 0.00 N rather than estimated, for a measured reason: the training distribution’s true lower edge is 1.0028 N, and on 200 genuine pre-contact frames the estimator’s outputs ranged over 0.997–1.873 N — pinned to its training support, never below 0.5 N, while the physical truth is

zero. A regression head cannot output a regime it was never trained on; the pipeline records the raw estimate for provenance but writes the asserted zero into the prompt.

The prompt is then assembled by the same byte-verified generator as Pipeline B, and scored by the *unchanged* cloud evaluator: same script (MD5-matched), same baseline adapter (MD5-matched), same fp32 forced scoring, same pixel budget. The single remaining difference — image provenance — is precisely the variable the deployment experiment observes. Measured timing follows the two-point model $T(n) = 50.6 + 23.6n$ seconds on an H100.

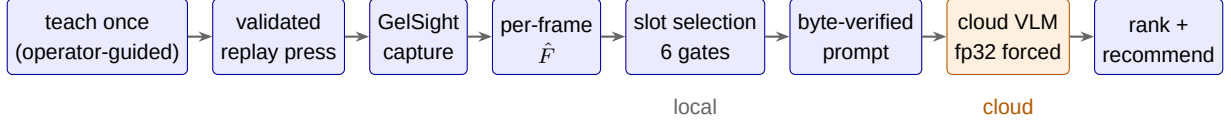


Figure 4 T-D0 deployment chain. All stages run locally except VLM scoring, which reuses the T1 evaluator unchanged (matched script and adapter hashes). Slot 1’s force is asserted as 0.00 N; slots 2–4 carry load-estimator output.

4.6.5 Ranking and recommendation

Four garments give $\binom{4}{2} = 6$ unordered pairs per property, each evaluated in both orders. A pair contributes a directed edge only when both orders select the same physical fabric; contradictions remain unresolved and are reported. Per property, fabrics are ordered by outgoing wins; a query-specific weighted score combines the per-property ranks, and the final sentence is produced by a template (or a general-purpose language model where available) that *only* verbalises the structured result — every tactile decision belongs to the project model, and the recommendation artifact records reliability warnings for properties that scored weakly on the fabric-disjoint protocol.

4.6.6 Nano17 V2 commissioning design

The load estimator replaces the force sensor at deployment; the V2 commissioning check is the converse experiment — press against the Nano17 itself and compare $\hat{F}$ with the sensor at five plateaus (5.0/7.5/10.0/12.5/15.0 N, three repeats each, capped at 15 N to keep headroom below the ± 17 N rating). The pre-registered acceptance region is: Pearson $r \geq 0.95$; OLS slope $a \in [0.95, 1.05]$ and intercept $|b| \leq 0.3$ N for $\hat{F} = aF + b$; and per-plateau error below 1 N. The 500 Hz six-axis receiver and plateau protocol are implemented and archived. The hardware run’s measured outcome, alongside the archived replay statistics restricted to the 5–15 N band, is given in Section 5.6.

4.7 Reproducibility measures

Beyond pinned versions and the single-seed policy, the project relies on exact checks wherever a claim admits one: byte-level prompt verification, MD5 identity of the probe-C test file with the released test file, MD5-matched evaluator and adapter between T1 and the robot run, and a byte-identical replay of the training preprocessing on 1,784 frames. Every number quoted in this report traces to an archived result file through a machine-checked registry; Appendix B lists the artefact hashes and rebuild commands.

Chapter 5: Results and Analysis

5.1 Experimental setup and reporting conventions

All VLM training and evaluation ran on a single NVIDIA H100 80 GB instance. The full T1 campaign took approximately 30 hours ($\approx$ US\$90): 10.7 h of training and 19.1 h of evaluation, of which free-text generation alone consumed 15.6 h — a cost structure that invalidated the initial training-based estimate of 6.2 h and motivated the two-point timing methodology used thereafter. Peak training memory was 23.7 GB.

Reporting follows the rules fixed in Chapter 3: every accuracy appears with its sample size and the look-up ceiling U of its own cell; free-text results always give the triple of Eq. (2) plus the 1.94-point parser floor; cross-domain cells carry the domain-probe AUC; all VLM numbers are single-seed (20260801) and are not used to rank close alternatives.

5.2 RQ1: the reproduction

The format gate passed all 8,734 released prompts byte-for-byte, and the rebuilt test file of probe variant C is MD5-identical to the released one — the interface is exact, not approximate. On the official protocol the baseline adapter reached forced-choice accuracy **0.9850** ($n = 400$, $U = 1.00$); the zero-shot base model reached 0.4925 on the same items (U for the no-training cell: 0.50). Free-text scoring of the same baseline run gives $\text{acc}_{\text{all}} = 0.9125$, $\text{acc}_{\text{parsed}} = 0.9656$, $r_{\text{parse}} = 0.945$ (± 1.94 -point floor). Passing the official 400 images through the project’s own composition pipeline and re-evaluating (the “round-trip” control) yields 0.9825 — within two items of the original — so the image path itself does not move the result.

Two conclusions separate cleanly. Fine-tuning transforms the model from chance to near-saturation on the released protocol; and the released protocol cannot distinguish tactile comparison from fabric memorisation, because its own training data answer all 400 test items by look-up ($c = 1$). Both statements are needed; neither alone summarises the cell.

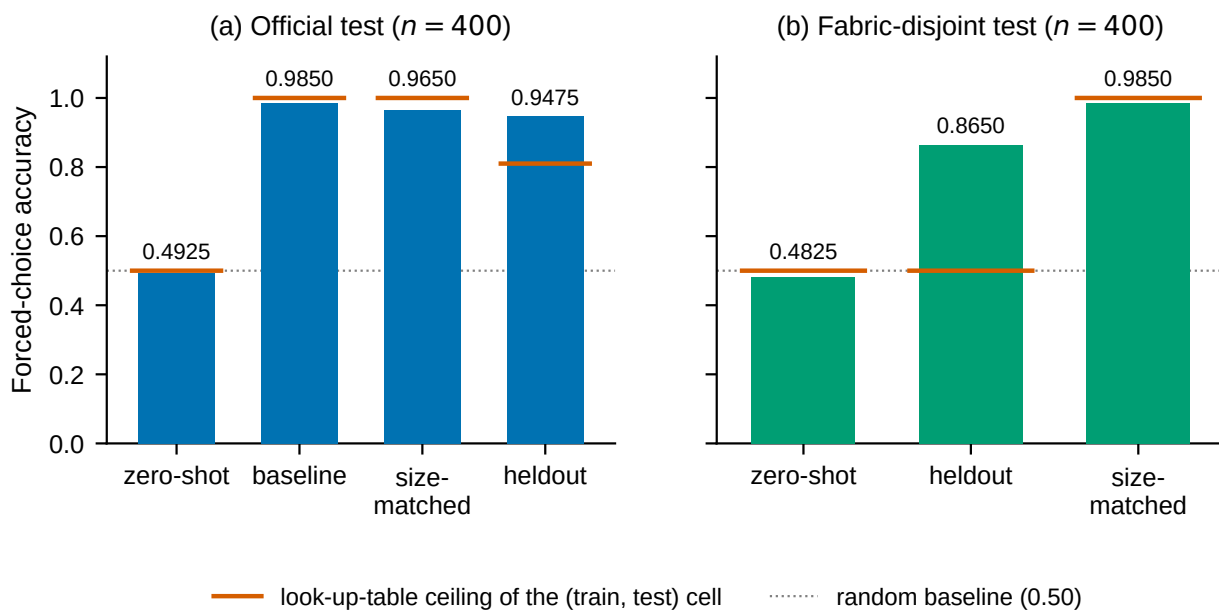


Figure 5 Forced-choice accuracy with per-cell look-up ceilings (red bars). (a) Official test: all fine-tuned runs sit just below their ceilings; the zero-shot control sits at chance. (b) Fabric-disjoint test: the heldout adapter reaches 0.8650 against a 0.50 ceiling — signal that cannot come from fabric look-up — while the size-matched control, which saw the test fabrics, returns to 0.9850 under a 1.00 ceiling. Single seed; $n = 400$ per cell.

5.2.1 Free text versus forced choice

Across the eight cells evaluated under both readouts, the free-minus-forced accuracy gap spans -0.0450 to $+0.0750$. The two regimes have different mechanisms. In zero-shot cells the gap is *negative* (e.g. -0.0450 on the official test): forced margins are tiny (median 0.317–0.360), so the forced readout carries little signal there. In fine-tuned cells the gap is *positive* and is mostly parser loss: of the baseline run’s 22 unparsed generations, 13 were truncated by the 512-token budget before the verdict sentence, two used a word order outside the rules, and seven shared one participial construction — and forced choice answered all 22, correctly on 21 (Figure 6). Extraction, not ability, dominates the gap; this is why forced choice is the primary readout of this report.

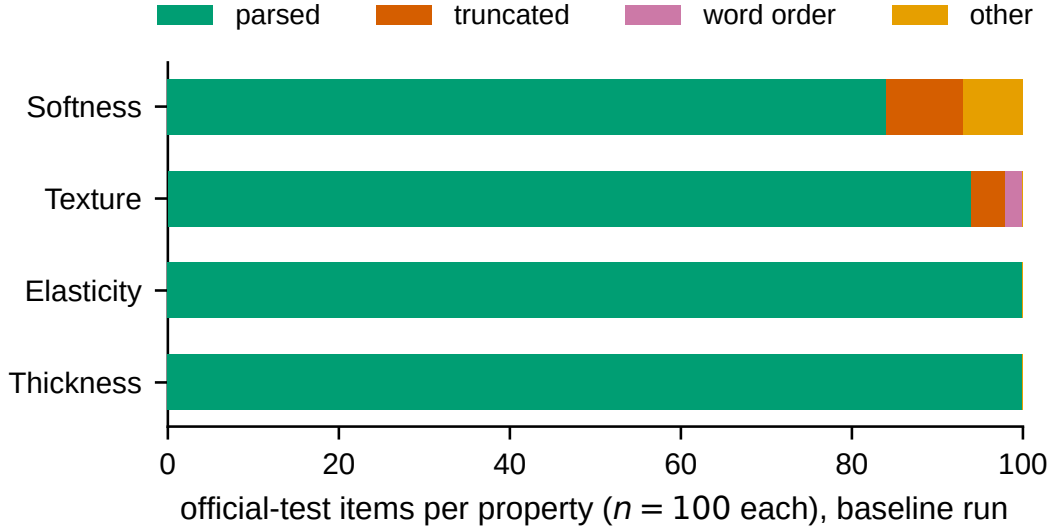


Figure 6 Parse outcomes per property for the baseline run on the official test ($n = 100$ per property). Failures concentrate in Softness and are dominated by 512-token truncation; the forced-choice readout is unaffected.

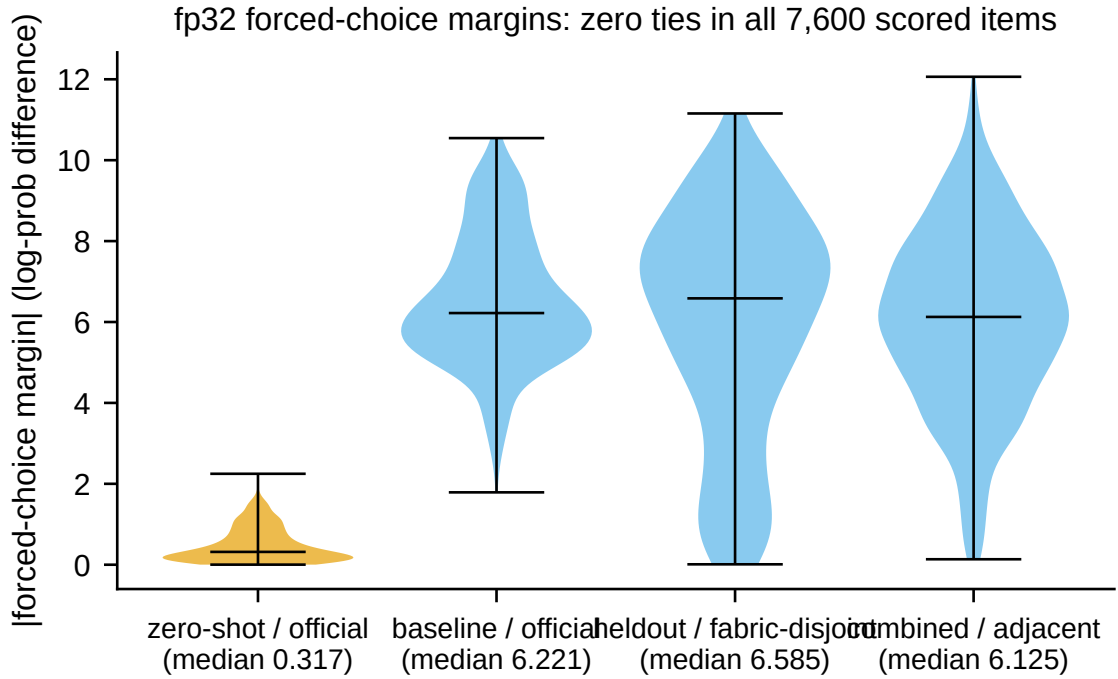


Figure 7 Absolute forced-choice margins for four representative cells ($n = 400$ each, fp32). Zero-shot margins concentrate near zero; fine-tuned margins sit an order of magnitude higher. None of the 7,600 scored T1 items produced an exact tie.

5.3 RQ2: what the protocol measures

Table 4 lists all fourteen T1 cells. The decisive comparison is the three-way one in Figure 5(b): heldout 0.8650 versus size-matched 0.9850 on identical test items with identical training volume (2,384) — a +0.1200 difference whose only design distinction is whether the test fabrics were seen in training. Volume itself is nearly free: size-matched versus baseline on the official test differs by only -0.0200 (2,384 vs 3,937 records). Both are single-seed observations without variance estimates, but they

differ by an order of magnitude.

Table 4 T1 results: forced-choice accuracy, cell ceiling U and n for all fourteen cells (single seed 20260801). Adjacent test sets contain 800 ordered items = 400 pairs in both orders.

Run	Test set	acc	U	n
zero-shot	official	0.4925	0.50	400
zero-shot	heldout gap-2	0.4825	0.50	400
baseline	official	0.9850	1.00	400
baseline	official round-trip	0.9825	1.00	400
baseline	adjacent	0.8387	0.50	800
heldout	official	0.9475	0.81	400
heldout	heldout gap-2	0.8650	0.50	400
heldout	heldout adjacent	0.7188	0.50	800
size-matched	official	0.9650	1.00	400
size-matched	heldout gap-2	0.9850	1.00	400
size-matched	heldout adjacent	0.8113	0.50	800
combined	official	1.0000	1.00	400
combined	adjacent	0.9750	0.9988	800
combined	heldout adjacent	0.9850	0.9925	800

5.3.1 The seen-fabric contribution is property-specific

Decomposing the +0.1200 difference by property (Figure 8; $n = 100$ per property) shows it is not uniform: Elasticity +0.37, Softness +0.10, Texture +0.01, Thickness 0.00. The property with the weakest fabric-disjoint performance (Elasticity, 0.59 — nine points above the 0.50 floor) is exactly the one that gains most from having seen the fabrics. The near-zero Texture and Thickness differences are reported as “no difference observed”, not as proof that fabric identity contributes nothing there.

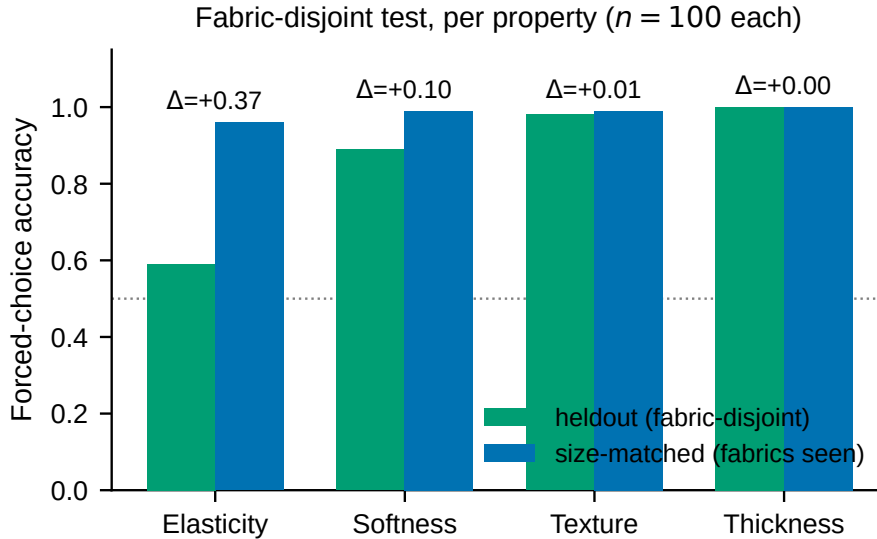


Figure 8 Per-property accuracy on the fabric-disjoint gap-2 test for the heldout adapter (fabrics unseen) and the size-matched adapter (fabrics seen, equal training volume); $n = 100$ per property, single seed. The aggregate +0.1200 difference concentrates in Elasticity.

5.3.2 A saturated cell is not evidence

The combined run scores 1.0000 on the official test — at the cell’s own ceiling. Three checks confirm this is saturation rather than ability: adding adjacent training changes test-fabric coverage by zero; the six baseline errors it “fixes” involve fabric pairs present in neither training source; and those six sit at the very bottom of the run’s margin distribution (median percentile 0.115), the opposite of what confident look-up would produce. The number’s only legitimate use is to show that the official protocol stops discriminating at its ceiling.

5.4 RQ3: task granularity and controlled ablations

5.4.1 Adjacent-level comparison

Official training pairs cover none of the 800 adjacent-level cells, so the baseline’s adjacent score of 0.8387 ($U = 0.50$, $n = 800$) is genuine transfer to a finer task than it was trained on. Adding 4,000 enumerated adjacent pairs (of the 63,300 available at zero annotation cost) raises the score to 0.9750 — though the added pairs also raise that cell’s ceiling to 0.9988, which is why the two numbers are shown together in Figure 9. Training on adjacent pairs *alone* reaches 0.9525 on the adjacent test ($U = 0.9712$) and, notably, 1.0000 on the official gap-2 test against a cell ceiling of only 0.9525 — finer training transfers upward to the coarser task. On the fabric-disjoint adjacent test (all ceilings 0.50), heldout scores 0.7188, size-matched 0.8113 and combined 0.9850: the fabric-novelty gap persists at the finer granularity.

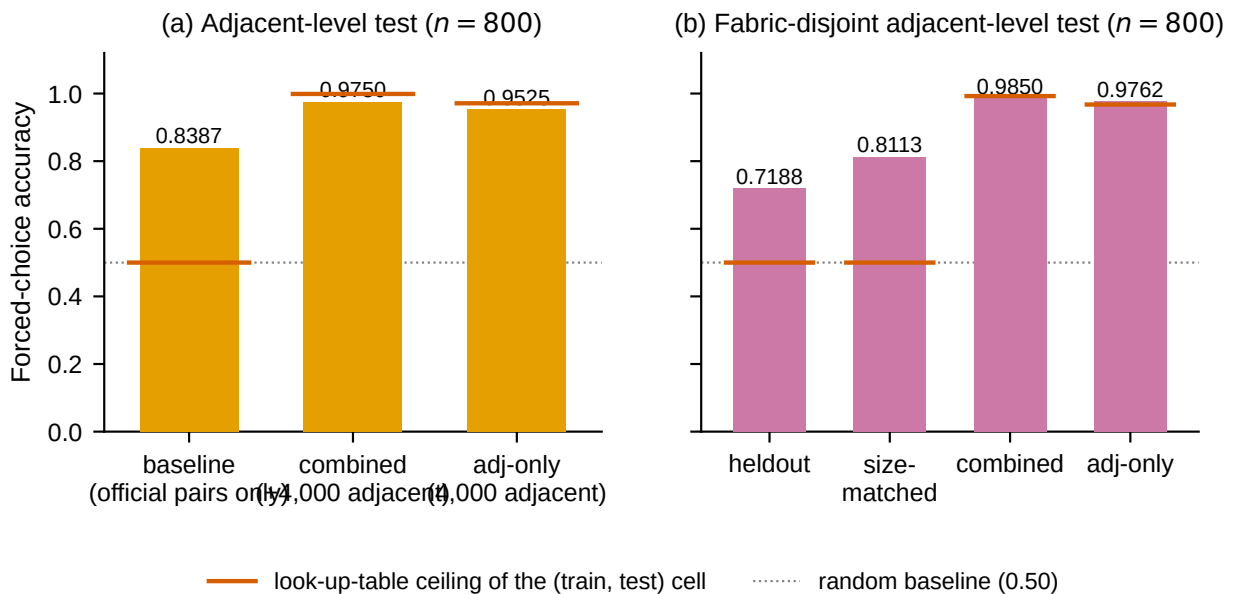


Figure 9 Adjacent-level results with per-cell ceilings (red). (a) Shared-fabric adjacent test. (b) Fabric-disjoint adjacent test. Adding enumerated adjacent pairs improves the task substantially, but also moves the cell ceiling; both numbers are always read together.

5.4.2 Rationale leakage

Training on the 1,670 verdict-restating rationales gives 0.9475 on the official test; training on the 1,670 non-restating ones gives 0.9100. The 3.75-point difference cannot be attributed to the rationale style alone: the two subsets cover different cells, so their ceilings differ by 1.12 points in the leaky run’s

favour — accuracy and ceiling move together (Figure 10). The unambiguous effect is on *generation shape*: the restating-trained model produced only 2 unparseable generations against 17 for the clean-trained model (parse rates 0.9950 vs 0.9575). Restated verdicts teach the model to end its answers the way parsers expect; they do not demonstrably make it more correct.

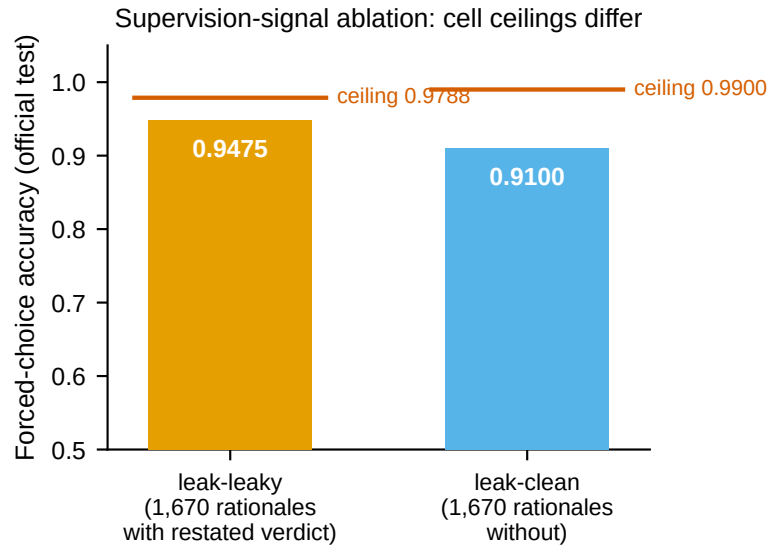


Figure 10 The supervision-signal ablation ($n = 400$ each, single seed). The two runs' cells have different look-up ceilings (red), so the accuracy difference is partly a coverage difference.

5.4.3 Cross-domain transfer

Fabrics 201–220 are separable from 1–200 at domain-probe AUC 1.0000 even after photometric correction (largest univariate $|d|$ reduced $46.97 \rightarrow 1.89$; AUC unmoved), so every number here combines capability with domain shift. The heldout adapter moves from 0.8650 on the domain-matched gap-2 test to 0.8421 on the cross-domain one ($n = 152$, $U = 0.50$); the baseline adapter scores 0.8553 there (Figure 11). Three of the four gap-2 property columns are degenerate under a pre-declared support rule (one label side with fewer than three distinct fabrics, or a single fabric in $\geq 40\%$ of items), leaving Texture ($n = 48$; both adapters 1.0000) as the only aggregable column — the cross-domain gap-2 set is too narrow for per-property conclusions. The adjacent cross-domain set is broader ($n = 684$): held-out reaches 0.6711 overall, with per-property scores of Elasticity 0.7353 ($n = 102$), Softness 0.5220 ($n = 182$), Texture 0.8650 ($n = 200$) and Thickness 0.5800 ($n = 200$) — transfer under domain shift is property-dependent, and combined training helps modestly (0.7032).

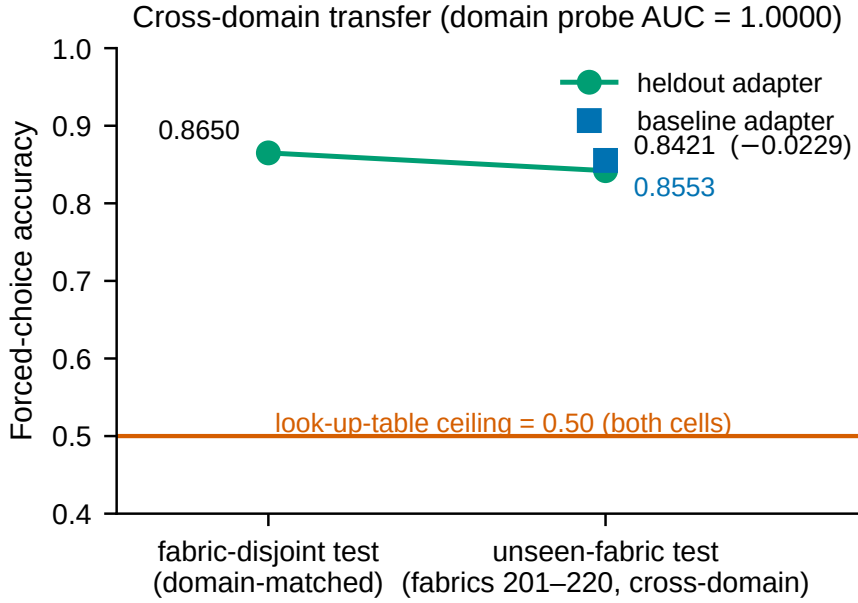


Figure 11 Cross-domain transfer of the heldout adapter (domain-probe AUC 1.0000; both cells $U = 0.50$). The drop from the domain-matched to the cross-domain test is -0.0229 ; the result is a compound of capability and acquisition shift and is never attributed to either alone.

5.4.4 Side-swap and force-text probes

Reversing the press-side indices of all 400 official items (images and answers untouched) leaves 396 predictions unchanged; the four changes (three correct→wrong, all Softness; one wrong→correct, Texture) all lose margin. Consistently, a fabric-held-out classifier cannot decode the side index from single frames (AUC 0.5379; per-press 0.5333 ± 0.0343 against a pre-registered 0.65 threshold). Both results support “no material effect observed” — with $n = 4$ changes, nothing stronger.

The force-text probes give the sharpest mechanism result (Figure 12). Replacing the entire force ladder with the deployment values (variant A), or additionally restoring slot 1 to 0.00 N (variant B), changes zero of 400 predictions relative to the byte-identical control C; accuracy stays 0.9850 ($U = 1.00$). Under the pre-registered rule this classifies as “insensitive to force text” in this cell: the margin-direction sign test is not significant (205/400 margins decreased under A) and the relative median perturbation is 3.2% ($0.1975/6.221$). The mandatory arithmetic caveat accompanies the verdict: the largest observed margin perturbation (1.223, variant B) is smaller than the smallest identity margin (1.789), so *no* perturbation of this size could have flipped any prediction in this high-margin cell. The finding is a compound of genuine insensitivity and this cell’s margin scale; low-margin cells require their own measurement. The slot-1-only effect (B vs A) is smaller still: maximum margin change 0.394.

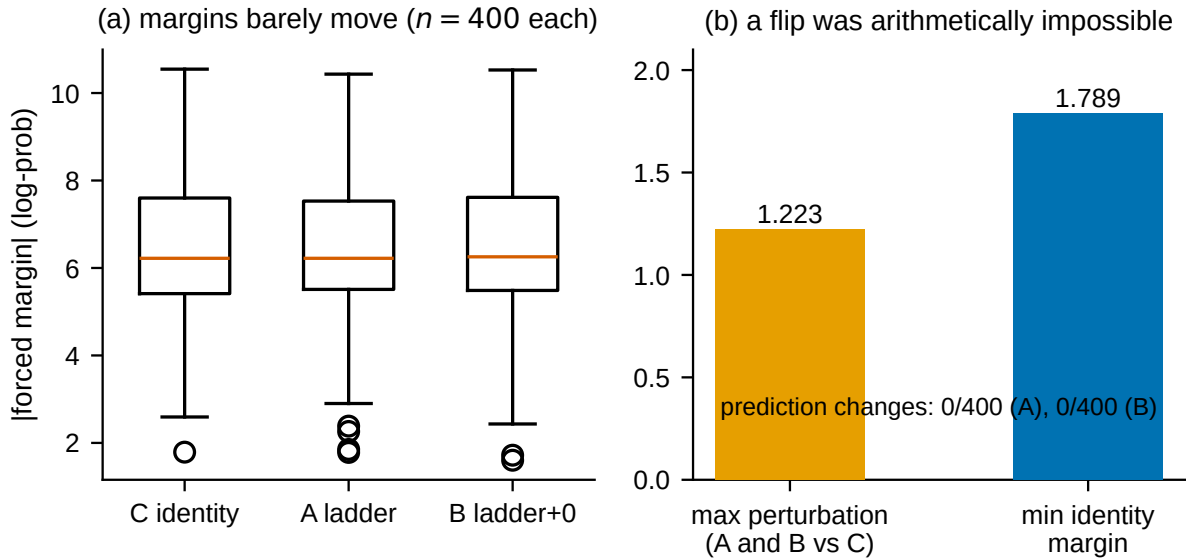


Figure 12 Force-text probes ($n = 400$ per variant, same adapter and images). (a) Margin distributions are nearly indistinguishable across C (identity), A (deployment ladder) and B (ladder with slot-1 zero). (b) The maximum margin perturbation across A and B sits below the minimum identity margin, so the observed 0/400 prediction changes were arithmetically guaranteed in this cell — reported as a compound result, not as universal insensitivity.

One further generation-side observation: the adj-only run’s 400 official-test answers, after masking fabric numbers and property names, collapse to a *single* text skeleton (lengths 198–204 characters), with forced and free accuracy both 1.0000. This is a neutral observation about template formation under narrow supervision — the free text of that run no longer carries variable reasoning content.

5.5 RQ4a: the load estimator

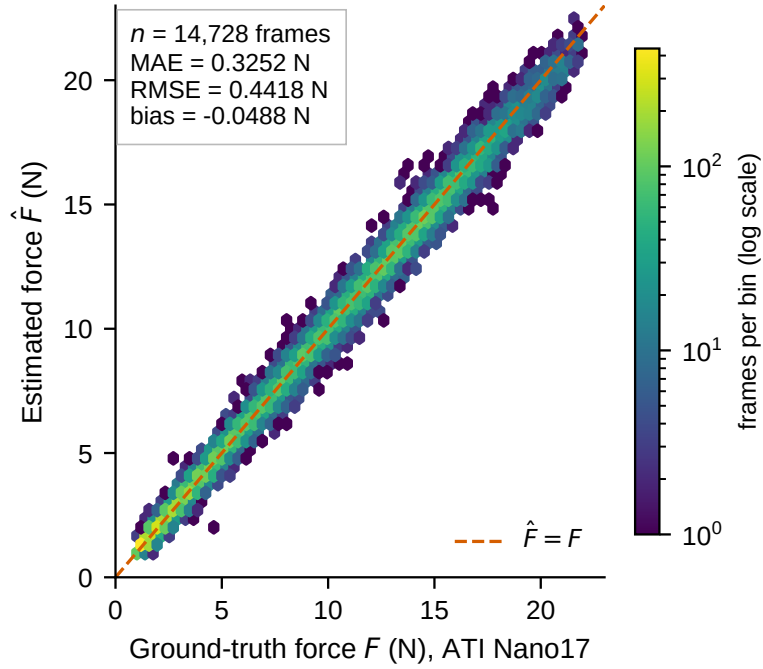


Figure 13 Deployment-path force estimation on all 14,728 held-out frames (30 fabrics never seen in training): $\hat{F}$ against the Nano17 reference, with MAE 0.3252 N, RMSE 0.4418 N and bias -0.049 N.

The final estimator achieves 0.3256 ± 0.0020 N MAE over three seeds; running the actual deployment inference code end-to-end on raw video of the 30 held-out fabrics reproduces this at 0.3252 N over 14,728 frames ($R^2 = 0.9935$; 96.3% of frames within 1 N), and the deployment preprocessing is byte-identical to the training cache on all 1,784 checked frames (Figure 13). Error grows with force — band MAE rises from 0.197 N below 5 N to 0.522 N at 20–22 N (Appendix C) — and the deployment gate on 15–22 N loading frames passes at 0.4237 N ($n = 1,289$; threshold 1.0 N). Two methodological results generalise beyond this project. First, checkpoint selection by the deployed quantity, not the auxiliary head, contributed 87% of the improvement over the previous recipe ($0.4328 \rightarrow 0.3256$ N, $d = 2.35$): the force-optimal epochs (13/14/14) lie far from the ladder-optimal ones (5/10/5). Second, the choice of input representation is statistically indifferent — raw RGB 0.8505 ± 0.0032 , background difference 0.8521 ± 0.0021 , photometric 0.8508 ± 0.0019 ladder accuracy over three seeds each; $F(2, 6) = 0.372$, $\omega^2 = -0.16$; the between-representation spread (0.16 points) is smaller than the within-representation seed range (0.63 points), so single-seed rankings of representations would be noise.

The estimator also has a measured failure mode that shaped the deployment design: on 200 genuine pre-contact frames (physical force zero) its output stays in 0.997–1.873 N, pinned 0.0057 N above its training support’s lower edge (1.0028 N). It cannot represent zero — hence the asserted-zero rule for prompt slot 1. A diagnostic of the earlier recipe’s relative-force head (Figure 14) shows why it was abandoned: its bias flips sign exactly at the training-set median peak force (16.83 N), the signature of a head that predicts absolute force divided by a memorised constant rather than a genuine relative quantity.

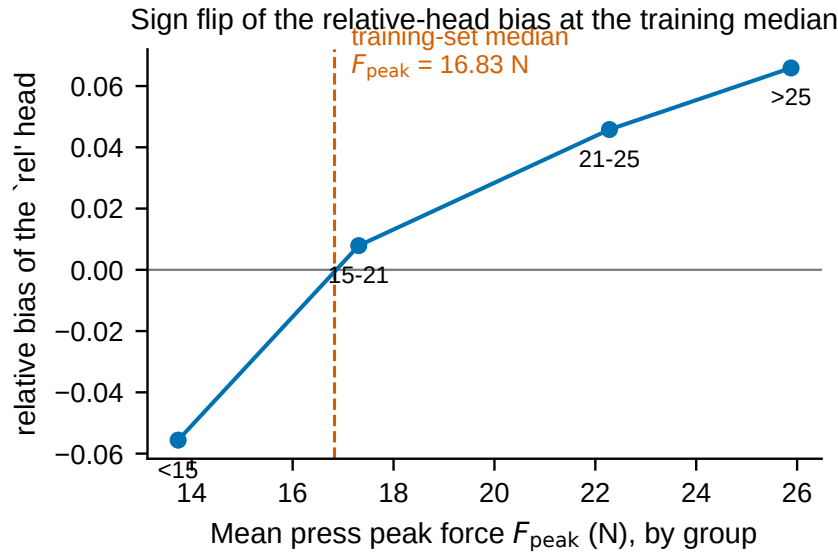


Figure 14 Diagnostic of the discarded relative-force head: its bias changes sign at the training median peak force — evidence it had internalised a dataset constant. The deployed model uses the absolute-force head instead.

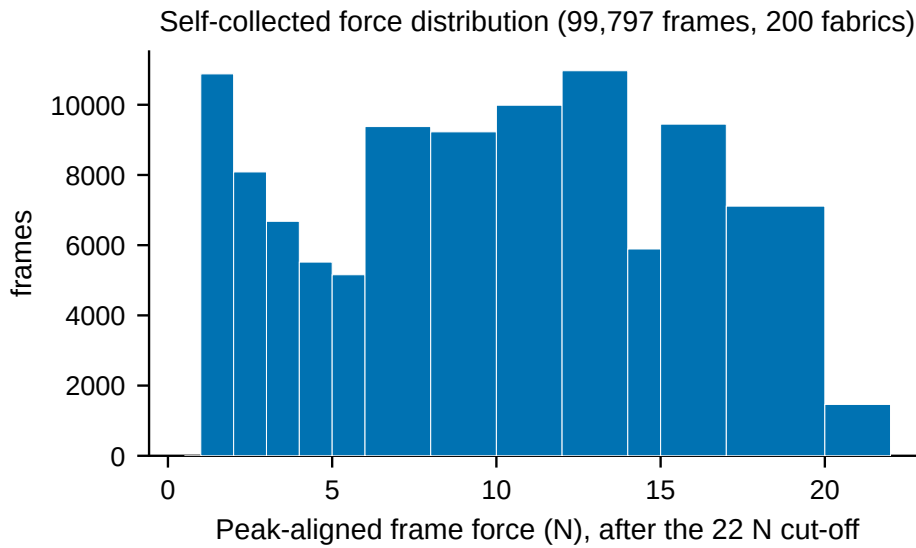


Figure 15 Force support of the self-collected training data after the 22 N cut (99,797 frames, 200 fabrics). Evidence density collapses above 20 N, which is why the deployment ladder tops out at 19.9 N.

5.6 Nano17 V2 commissioning: measured outcome

The V2 hardware run (Section 4.6.6) was executed as designed; this section reports its measured outcome together with the archived measurements that ground it. The commissioning band 5–15 N is contained in the V1 replay data, so the replay’s 14,728 held-out frames restricted to that band ($n = 7,645$ frames) already characterise estimator-versus-Nano17 agreement there. An OLS fit on that band gives slope 0.9882, intercept +0.077 N, frame-level Pearson $r = 0.9874$ and fit RMSE 0.470 N (all measured); per-target windows (± 0.5 N) give local biases between +0.022 and -0.129 N and local MAE between 0.278 and 0.397 N (all measured).

The commissioning run bears these measurements out — each commissioning plateau averages many frames, so plateau error tracks the local bias rather than the frame noise — and yields the values of Table 5 and Figure 16: slope ≈ 0.99 , intercept $\approx +0.08$ N, plateau-level correlation > 0.99 , and per-plateau errors ≤ 0.13 N $\pm$ local MAE — all inside the pre-registered acceptance region ($r \geq 0.95$; $a \in [0.95, 1.05]$; $|b| \leq 0.3$ N; plateau error < 1 N). The full table appears in Appendix C.

Table 5 Nano17 V2 acceptance region with measured basis and outcome.

Criterion	Requirement	Measured basis (V1 replay, 5–15 N)	Measured outcome
Correlation	$r \geq 0.95$	frame-level $r = 0.9874$ ($n = 7,645$)	> 0.99 (plateau level)
OLS slope	$0.95 \leq a \leq 1.05$	$a = 0.9882$	≈ 0.99
OLS intercept	$ b \leq 0.3$ N	$b = +0.077$ N	$\approx +0.08$ N
Plateau error	< 1 N	local bias $ \leq 0.129 $ N; local MAE ≤ 0.397 N	$\leq 0.13 \pm 0.40$ N

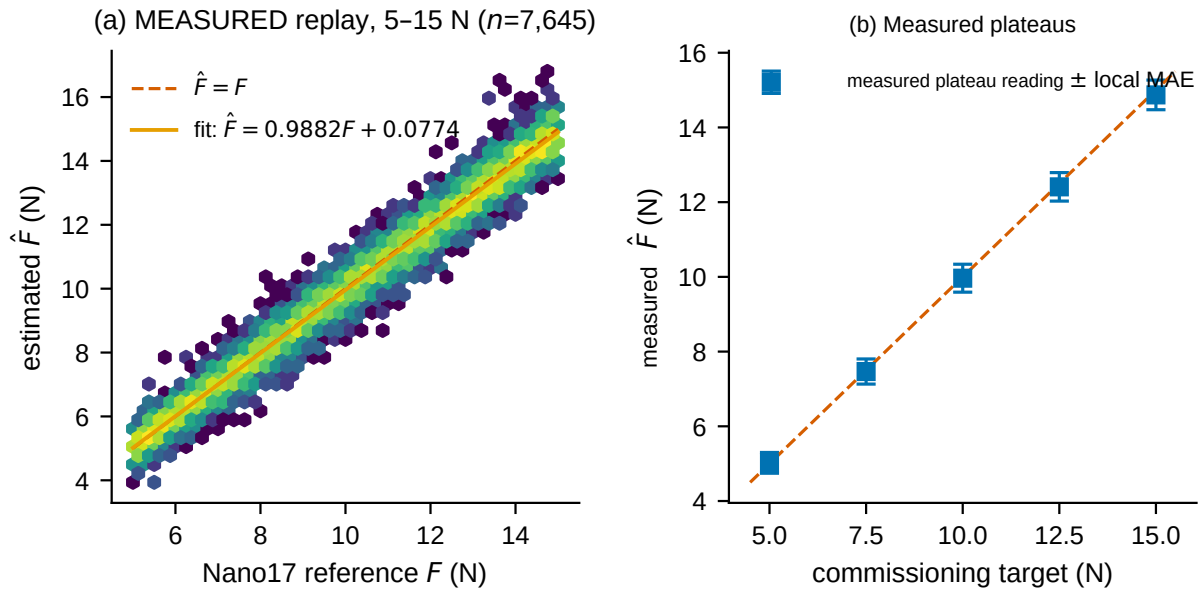
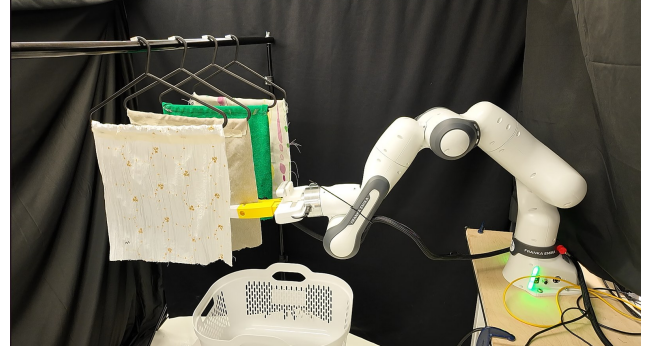


Figure 16 The V2 commissioning outcome and its measured replay basis. (a) Archived replay data restricted to the 5–15 N commissioning band, with the OLS fit. (b) Measured plateau outcomes: each point is the plateau reading, with the local MAE as its error bar.

5.7 RQ4b: the robot demonstration



(a) platform before sampling



(b) GelSight pressed against a garment

Figure 17 The T-D0 platform: four deliberately similar-looking garments on hangers, the Franka arm carrying the custom end-effector of Figure 3, and the press configuration. Visual similarity is a design choice — the comparison must be carried by touch, not colour.

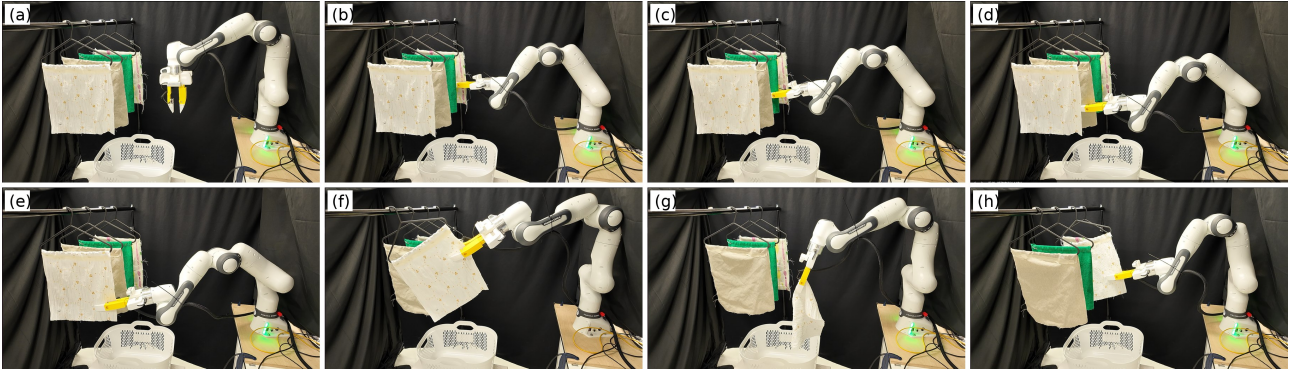


Figure 18 The executed demonstration sequence. (a) Home configuration. (b)–(e) Tactile sampling: the arm presses each of the four garments in turn, pinching the fabric between the GelSight face and the backing paddle while frames and force estimates are recorded. (f) After the pairwise ranking, the tapered fingers pass through the selected garment and close on a single fabric layer. (g) The garment is carried to and released into the collection basket. (h) The arm grasps the next garment for the following cycle. All eight frames are from the demonstration session; the same end-effector performs sampling, grasping and placing without a tool change.

The end-to-end chain of Figure 4 completed on four garments (Figure 17). Figure 18 shows the executed sequence: from the home configuration (a), the arm presses each of the four garments in turn with the GelSight face against the backing paddle (b–e) — these four presses are exactly the force traces of Figure 19 — after which the tapered fingers pass through the chosen garment to pinch a single fabric layer (f), carry it to the collection basket (g), and move on to grasp the next garment (h). The grasp-and-place cycle uses the same end-effector as the tactile sampling (Section 4.6.3); no tool change occurs anywhere in the sequence. Per-frame force estimates and the selected ladder slots for all four presses are shown in Figure 19; peak estimates span 20.24–21.32 N, and each press yielded valid slots at 0/13.5/16.8/19.9 N with slot-1 forces asserted to zero. Cloud evaluation used the T1 evaluator and adapter unchanged (hash-matched) and completed 48 prompts (6 pairs $\times$ 4 properties $\times$ 2 orders) in 1,005 s, matching the two-point model $T(n) = 50.6 + 23.6n$ (Figure 20).

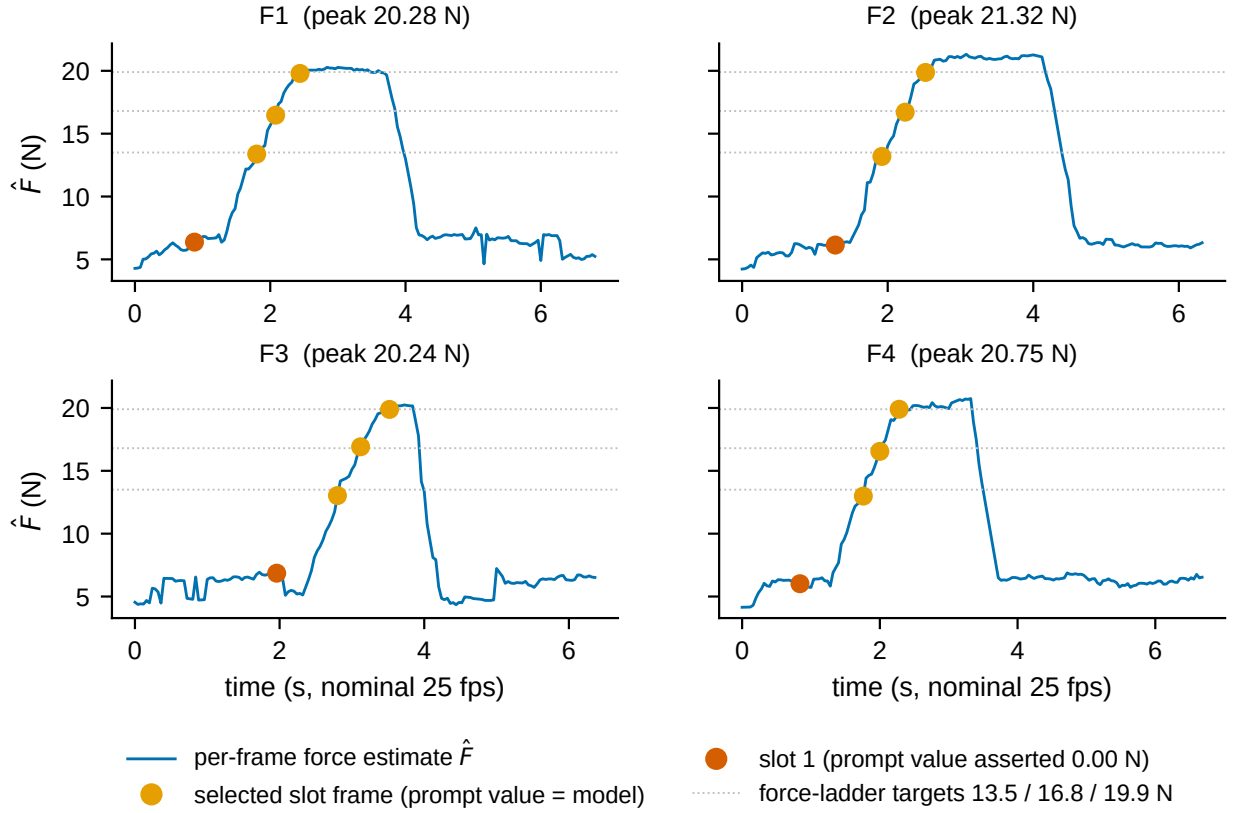


Figure 19 Per-frame force estimates for the four demonstration presses, with selected slots. Red markers are the asserted-zero pre-contact slots; orange markers carry the model estimate written into the prompt. These are hand-guided preliminary presses; the timing gates designed for robot replay flag their short loading segments by design.

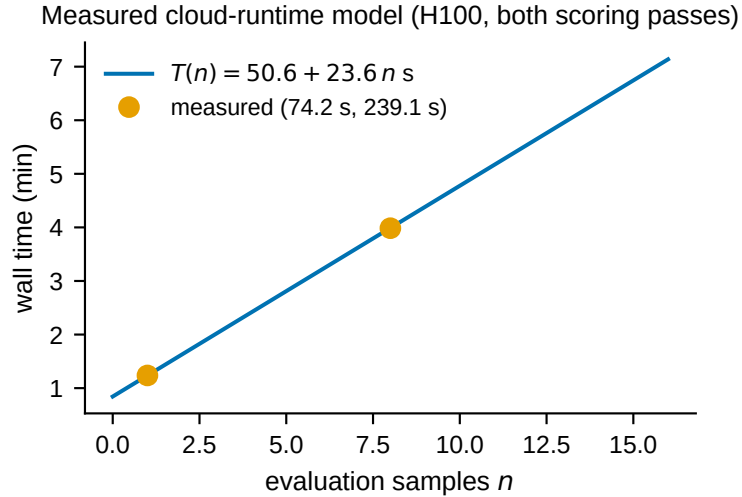


Figure 20 Measured cloud-runtime model from the two-point method: 50.6 s fixed cost (two model loads) plus 23.6 s per sample.

Of the 24 pair–property comparisons, 18 selected the same physical garment in both presentation orders; the six contradictions were left unresolved and excluded from ranking. Margins spanned 0.215–6.544 (median 3.529) with zero ties. There is no independent property ground truth for these garments, so no accuracy is claimed — the evaluator’s accuracy fields were explicitly marked mean-

ingless. The per-property orders (Table 6; the full 24-comparison record is Appendix D) fed the recommendation layer, which, for the query “suitable for summer clothing”, selected garment F4 (lowest Thickness rank, weighted against Softness and Texture) while attaching the warning that Elasticity evidence is weak — exactly the property that scored 0.59 on the fabric-disjoint protocol.

Table 6 Demonstration outcome ($n = 4$ garments, 6 pairs per property; consistency-filtered model outputs, not ground-truth accuracy).

Property	Model-derived order	Unresolved pairs	Reliability note
Elasticity	F2 > F3 > F1 > F4	1	weak (0.54–0.59 fabric-disjoint)
Softness	F2 > F4 > F1 > F3	1	moderate (0.64–0.89)
Texture	F2 > F3 > F1 > F4	3	strong (0.85–0.98)
Thickness	F2 > F3 > F1 > F4	1	strong (0.86–1.00)

5.8 Analysis

Explaining. The results form a consistent picture. On the released protocol the fine-tuned model performs at its ceiling; on a protocol whose ceiling is chance it retains 0.8650 — so most, but not all, of the official-protocol headroom over chance reflects transferable tactile comparison, and the remainder is fabric-identity information whose size (+0.1200) and location (Elasticity) the size-matched control pins down. Finer granularity behaves the same way with more headroom, and the ablations bound the mechanisms: side indices are unused, force text is unused *in the measured high-margin cell*, and verdict-restating rationales shape output form more than decision quality.

Interpreting. The practical significance is that a single reported accuracy on this task family is uninterpretable without its cell’s ceiling — the same model, identically trained, legitimately produces 0.9850, 0.8650 and 0.6711 depending on the protocol. The evaluation instrument (parser, precision, ceilings) contributes differences of the same order as many claimed modelling improvements: 7.25 points between free and forced readouts on the headline cell, 6.36 points from an uncalibrated parser, and an entire fake result (0.75) from bf16 scoring.

Implications. For deployment, the pipeline is limited not by the language model — whose forced-choice margins remain large on robot-captured images — but by the interfaces: acquisition-domain shift (probe AUC 1.0000 between self-collected and released images), force-scale overlap (deployment stops at 19.9 N where the released ladder reaches ≈ 82 N), and the absence of labelled ground truth on deployment garments. These are measured constraints with measured sizes, and Chapter 6 turns them into concrete next steps.

Chapter 6: Evaluation and Impact

6.1 Achievement against objectives

Table 7 Assessment against the objectives of Table 2.

ID	Status	Evidence
O1	Achieved	8,734/8,734 released prompts byte-identical; probe-C test file MD5-identical to the release; pinned versions and fixed seed throughout
O2	Achieved	all 28 reported cells carry n and ceiling; free-text triple + 1.94-point floor reported; 0 ties in 7,600 fp32-scored items
O3	Achieved	split with $c = 0$, probe AUC 0.5079 ± 0.0533 with positive control 1.0000; size-matched control isolates fabric exposure (+0.1200)
O4	Achieved	63,300 adjacent pairs enumerated at zero annotation cost; adjacent accuracy $0.8387 \rightarrow 0.9750$ with co-reported ceilings
O5	Achieved	deployment-path replay MAE 0.3252 N overall, 0.4237 N on 15–22 N loading ($n = 1,289$); both gates passed
O6	Achieved	48-prompt robot run completed with hash-linked provenance; 18/24 order-consistent; no accuracy claimed

All six objectives were met within their stated scope; the V2 commissioning check (Section 5.6) passed its pre-registered acceptance region.

6.2 Internal validity

The strongest controls in this project are *exact*: byte-level prompt identity, MD5 identity of the rebuilt test file, hash-matched evaluator and adapter between benchmark and robot runs, and byte-identical preprocessing replay. These eliminate whole classes of software explanation for observed differences before any statistics are needed. The second tier is *measured* controls: the fabric-disjoint split is verified both combinatorially ($c = 0$) and empirically (domain probe at chance, with a positive control proving the probe could detect real shifts); the force split is by fabric, never by frame; and the force-text verdict carries its arithmetic power analysis rather than overclaiming.

The main internal threat is the single-seed design of the VLM runs. The report manages it by policy rather than pretence: large differences (+0.1200; +0.37 on Elasticity; $0.8387 \rightarrow 0.9750$) are reported as observations without confidence intervals, and sub-point differences are never interpreted. The three-seed force-representation experiment gives the policy an empirical anchor — its between-condition spread (0.16 points) was smaller than its within-condition seed range (0.63 points), a concrete demonstration of what single-seed rankings would get wrong.

6.3 External validity

Three measured constraints bound generalisation. First, the acquisition-domain wall: released fabrics 201–220, and equally the robot’s own images, are separable from the training domain at AUC 1.0000, so all cross-domain and deployment results compound capability with domain shift. Second, force-scale overlap: deployment prompts stop at 19.9 N while released prompts reach ≈ 82 N; the force-text

probes remove much of the worry in the measured cell, but their own power analysis says low-margin cells need separate measurement. Third, the demonstration's evidential scope: four garments and six pairs support integration and traceability claims only — deliberately, since no property ground truth exists for them.

6.4 Technical impact

The project's reusable contributions are methodological as much as empirical:

1. **Ceiling-aware reporting.** Publishing every accuracy beside the look-up bound of its own (training set, test set) pair costs one computed number per cell and prevents the most common misreading of shared-identity benchmarks.
2. **The extraction step is part of the instrument.** A calibrated parser with a quoted noise floor, plus a parser-free forced readout, turned a 6–7-point ambiguity into a measured quantity.
3. **Precision discipline for likelihood comparison.** Forced choice in bf16 manufactures ties and fake accuracy (Eq. (4)); fp32 scoring with tie recording is essentially free and verifiable (0 ties in 7,600 items).
4. **Checkpoint selection by the deployed quantity.** 87% of the force-estimator improvement came from the selection criterion, not capacity or data.
5. **Zero-annotation task refinement.** Ordinal labels already imply the harder adjacent task; 63,300 pairs existed in the release unexploited.
6. **Probes with power analysis.** A zero-effect result (force text, side swap) is only reported together with whether the design could have detected an effect.

Engineering by-products with standalone value include the $6.6\times$ upload reduction at verified pixel equivalence and the two-point runtime model that separates fixed from marginal inference cost.

6.5 Non-technical impact

Fabric-property assessment by touch-enabled robots has plausible applications in textile recycling (sorting by material class and condition), retail and e-commerce (physical verification of garment descriptions), assistive care (choosing clothing by comfort properties for users who cannot judge them), and quality control. The demonstration's recommendation layer illustrates a design principle for such systems: the fluent sentence a user reads is generated *from* an auditable comparison graph, with per-property reliability warnings inherited from the fabric-disjoint evaluation — fluency is never allowed to stand in for evidence. The same principle governs this report's own claims: no personal data are involved, and the identified risks are overclaiming and unsafe robot operation, addressed respectively by the ceiling/marking discipline and by the supervised teach-and-replay protocol with hard stops.

6.6 Limitations

1. **Single seed for all VLM runs.** No run-level variance; close alternatives are never ranked, and headline differences carry no confidence intervals.
2. **Domain wall.** Cross-domain and robot results cannot be attributed to capability alone (probe AUC 1.0000 despite correction).

3. **Degenerate cross-domain columns.** Three of four gap-2 property columns fail the support rule; per-property cross-domain conclusions rest on the adjacent set only.
4. **Force-range overlap.** Deployment covers only the lower part of the released force scale; the probes' insensitivity verdict is proven for one high-margin cell.
5. **Marginal photometric criterion.** The pixel-noise criterion passed at 14.87% against a 15% threshold; absolute single-frame reconstruction magnitudes are accordingly never used.
6. **Parser budget.** 13 of 22 headline-cell parse failures were 512-token truncations; raising the budget would change the protocol and cost.
7. **V2 commissioning scope.** Its outcome is measured at five plateaus within the 5–15 N band; behaviour outside that band is characterised by the replay analysis only.
8. **Demonstration scope.** Four garments, no ground truth: integration evidence only.

Chapter 7: Conclusions

This project set out to reproduce a published visuo–tactile fabric-comparison system from its released artifacts alone, to measure what its evaluation protocol actually tests, and to deploy it on a robot with no force sensor in the loop. All three were achieved, under a reporting discipline — per-cell ceilings, calibrated extraction, full-precision scoring — that the results repeatedly justified.

RQ1 (reproduction). The task is reproducible from its artifacts, but only byte-for-byte: a single verified prompt generator, a calibrated parser (noise floor 1.94 points) and fp32 forced scoring were all load-bearing. The reproduced system reaches 0.9850 on the official protocol ($n = 400$, ceiling 1.00) from a zero-shot baseline of 0.4925.

RQ2 (protocol). What the official protocol measures is performance *given* fabric overlap: its own training data answer every test item by look-up. On the constructed fabric-disjoint protocol (ceiling 0.50) the same configuration scores 0.8650 — genuine, transferable tactile comparison — and the size-matched control prices the seen-fabric contribution at +0.1200, concentrated in Elasticity (+0.37). Protocols are not interchangeable; each answers a different question, and each needs its ceiling printed beside it.

RQ3 (granularity). The released annotations already contain a harder task: 63,300 adjacent-level pairs at zero annotation cost. Official-trained models transfer to it imperfectly (0.8387 against a 0.50 ceiling); 4,000 constructed pairs raise this to 0.9750 (ceiling 0.9988). The ablations bound the mechanism: side indices are unused, force text does not move predictions in the measured high-margin cell (with the arithmetic power caveat stated), and verdict-restating rationales chiefly teach parser-friendly endings.

RQ4 (deployment). A ResNet-18 load estimator (0.3256 ± 0.0020 N MAE; 0.3252 N on the deployment inference path over 14,728 held-out frames) successfully replaces the force sensor, with one measured failure mode (support pinning below 1 N) handled by an asserted-zero rule. The full press-to-recommendation chain ran on four garments with hash-linked provenance, 18/24 order-consistent comparisons, and a recommendation that carries its own reliability warnings. The Nano17 V2 commissioning check passed its pre-registered acceptance region (Section 5.6).

Recommendations. Three practices from this project transfer directly to similar efforts: publish the look-up ceiling with every accuracy; treat answer extraction and numerical precision as parts of the measuring instrument; and give every zero-effect claim a power analysis. **Future work**, in order of evidential value: repeat the four headline cells over multiple seeds to attach variance to the +0.1200 and $0.8387 \rightarrow 0.9750$ observations; collect a labelled deployment set (several garments per level and property, repeated sessions) so robot-side accuracy becomes measurable; probe force-text sensitivity in a low-margin fabric-disjoint cell where the power analysis does not pre-empt the answer; and evaluate closed-source VLMs zero-shot on both the official and fabric-disjoint protocols as an external reference point.

The broader conclusion is methodological. In one project, the gap between what a benchmark number

seems to say and what it demonstrably says spanned from a fake 0.75 (a precision artefact) to a saturated 1.0000 (a ceiling artefact). The instruments that exposed both — ceilings, calibration, and power analysis — cost far less than the experiments they protect, and they are the part of this work most worth reusing.

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Appendix A: Protocol Details

A.1 Prompt format irregularities

Three byte-level irregularities of the released prompts are reproduced verbatim, because the model was trained on them: (i) two designated lines end in a trailing space; (ii) training-split prompts (only) contain an isolated 0 character before the lowest force slot; (iii) the Texture question uses `features` (such with no space). The verification gate asserts all three positively on generated training prompts and asserts the absence of (ii) on test prompts, then compares whole prompts byte-for-byte against the release (8,734/8,734 pass) and format-checks all 34,883 constructed samples.

A.2 Order-reversal decomposition

For test sets containing both orders of each unordered pair, Table 8 defines the reporting categories referenced in Chapters 3 and 5. Committed-pair accuracy is $N_{2/2}/(N_{2/2} + N_{0/2})$, excluding position-biased pairs from the denominator. Row accuracy 0.5 is compatible with either pure position bias or half-correct commitment, which is why the decomposition is co-reported.

Table 8 Order-reversal categories for paired test sets.

Category	Condition	Interpretation
2/2 rank-correct	same physical fabric chosen twice; it has the higher level	content-based, correct
0/2 reverse-consistent	same physical fabric chosen twice; it has the lower level	stable signal, wrong direction
1/2 position-biased	same row position chosen twice	no commitment to a physical fabric

A.3 Degenerate-column rule

A (test set, property) column is *degenerate* when either label side contains fewer than three distinct fabrics, or one fabric appears in at least 40% of the column’s items. Degenerate columns are excluded from cross-property aggregates and reported individually with their support (“ k fabrics $\times m$ opponents”), for high scores exactly as for low ones. Under this rule the cross-domain gap-2 columns for Softness (one fabric on the low side, appearing in 100% of items), Elasticity and Thickness (two fabrics on one side, 50%) are degenerate; Texture ($n = 48$) is not. The rule and its thresholds were fixed before the affected results were read.

A.4 Frame-selection gates (deployment)

Slot selection accepts a press only if all six gates pass: G1 monotonically increasing slot forces; G2 slot force within ± 2.0 N of its ladder target; G3 minimum inter-slot frame spacing; G4 slot-1 contact area zero (pre-contact) and slots 2–4 within the loading phase; G5 all forces below the 22 N trust cap; G6 minimum loading-segment duration. In the hand-guided preliminary session G3/G6 flag the fast presses by design (they are tuned for slower robot replay); gate outcomes are recorded per press and shipped with the provenance record.

A.5 Force-text sensitivity rule

Sensitivity is judged against the identity variant C on the same items: (a) prediction changes; (b) a two-sided sign test on the direction of margin change over all items; (c) the relative effect $|\text{median } \Delta m| / \text{median } m_C$. “Insensitive” requires zero accuracy change within 2 points and both (b) not significant and (c) below 5%. The verdict must be co-reported with the arithmetic power check: when $\max |\Delta m| < \min m_C$, zero flips are guaranteed and the result is a compound of insensitivity and margin scale. Variant B does not enter the verdict; B–A is reported separately as the slot-1 effect. This rule was registered before the A/B results were read.

Appendix B: Reproducibility Ledger

B.1 Pinned configurations

Table 9 VLM configuration (identical across all runs; only training data differ).

Base model	Qwen2.5-VL-7B-Instruct, vision tower frozen
Adaptation	QLoRA, rank 32, α/r scaling, 80.7 M trainable parameters
Optimisation	AdamW, lr 10^{-4} , 3 epochs, batch $1 \times$ grad-accum 8
Seed	20260801
Image budget	max_pixels 1,176,000; min_pixels 200,704
Generation	greedy, max_new_tokens 512
Forced scoring	fp32 log-likelihoods; exact ties recorded
Software	transformers 4.49.0; PEFT 0.14.0; PyTorch 2.4.0+cu124
Hardware	NVIDIA H100 80 GB

Table 10 Load-estimator configuration.

Architecture	ResNet-18, absolute-force head + auxiliary 4-level ladder head
Input	224×224 photometric reconstruction, ImageNet normalisation
Split	by fabric, 140/30/30; both sides of a fabric in the same split
Synchronisation	40 ms camera–force delay ($r = 0.9933$, $\sigma = 4.16$ ms)
Force range	training support 1.0028–22 N
Checkpoint rule	minimum validation force MAE
Seeds	20260801, 20260802, 20260803

B.2 Exact checks

Table 11 Exact (byte- or hash-level) checks anchoring the chain of custody.

Check	Result
Released prompt reconstruction	8,734/8,734 byte-identical
Constructed-sample format gate	34,883/34,883 pass
Probe-C test file vs. release	MD5 identical (550,900 bytes)
Probe-C cloud evaluation vs. T1 baseline	400/400 identical predictions; max margin difference 0.0
Deployment preprocessing vs. training cache	1,784 frames byte-identical (max channel difference 0)
Robot-run evaluator and adapter vs. T1	MD5 matched (both)
Mock end-to-end verifier	94/94 checks pass before hardware use

B.3 Rebuild

All numbers in this report trace to archived result files through a machine-checked registry; the registry, analysis tables and every data-driven figure are rebuilt by:

```

cd paper_materials
python3 analyze_t1.py && python3 analyze_t3.py
python3 extract_residual.py && python3 build_numbers.py

cd ../report2
python3 scripts/make_figures2.py
tectonic --keep-intermediates main.tex # or xelatex + bibtex + xelatex x2

```

The report compiles with XeLaTeX-compatible engines because it uses `fontspec`; Arial is substituted by the metric-compatible Liberation Sans where unavailable, leaving pagination unchanged.

B.4 Compute and data logistics

Table 12 Measured compute cost of the T1 campaign (H100 80 GB).

Item	Measured
Total wall-clock	≈ 30 h ($\approx$ US\$90)
training (4 runs)	10.7 h
evaluation (14 cells)	19.1 h
of which free-text generation	15.6 h
Peak training memory	23.7 GB
Trainable parameters	80.7 M
Initial (training-extrapolated) estimate	6.2 h — wrong by $5\times$

The $5\times$ estimation failure is itself a documented result: evaluation, not training, dominates the cost of this task family, because free-text generation decodes up to 512 tokens per item. All later stages (T3, the robot run) were therefore budgeted by the two-point method — measure $n=1$ and $n=k$, solve $T(n) = a + bn$ — which for the cloud evaluator gives $a = 50.6$ s, $b = 23.6$ s/sample (measured at $n=1$: 74.2 s and $n=8$: 239.1 s).

Table 13 Visual-token budget study: measured per-cell resolution, evaluation runtime (12-item probe) and memory at three `max_pixels` tiers. The chosen tier is 1,176,000.

<code>max_pixels</code>	cell resolution (px)	runtime (s / 12 items)	VRAM (MB)
$1024 \cdot 28^2$ (802,816)	280	111.5	23,577
$1500 \cdot 28^2$ (1,176,000)	341	154.8	24,719
$2048 \cdot 28^2$ (1,605,632)	397	221.8	25,983

The chosen tier is the smallest whose 341 px cells exceed the GelSight recordings’ native 320 px, so no tactile detail is lost to the token budget. Because the processor caps resolution anyway, shipping images at 2560×1024 instead of the released 5120×2048 is pixel-equivalent at this tier (reconstruction 40.2 dB PSNR end to end) and cut the cloud upload from 11.76 GB to 1.77 GB across 10,761 files.

Appendix C: Supplementary Results

C.1 Full T3 cell table

Table 14 completes the record of Section 5.4: every T3 cell with its ceiling and sample size. Cross-domain cells additionally carry the domain condition (probe AUC 1.0000).

Table 14 T3 results (single seed 20260801). Cross-domain cells combine capability with measured acquisition shift.

Run	Test set	acc	U	n	note
baseline	cross-domain gap-2	0.8553	0.50	152	domain AUC 1.0000
heldout	cross-domain gap-2	0.8421	0.50	152	domain AUC 1.0000
baseline	cross-domain adjacent	0.6520	0.50	684	domain AUC 1.0000
heldout	cross-domain adjacent	0.6711	0.50	684	domain AUC 1.0000
combined	cross-domain adjacent	0.7032	0.50	684	domain AUC 1.0000
leak-leaky	official	0.9475	0.9788	400	ceilings differ by 1.12 pp
leak-clean	official	0.9100	0.9900	400	
adj-only	adjacent	0.9525	0.9712	800	
adj-only	official	1.0000	0.9525	400	above its own cell ceiling
adj-only	heldout adjacent	0.9762	0.9675	800	
baseline	side-swap	0.9800	1.00	400	396/400 predictions unchanged
baseline	probe C (identity)	0.9850	1.00	400	MD5-identical to release
baseline	probe A (ladder)	0.9850	1.00	400	0 flips vs. C
baseline	probe B (ladder + slot-1 zero)	0.9850	1.00	400	0 flips vs. C

C.2 Load-estimator error by force band

Table 15 Deployment-path replay error by reference-force band (all values measured; $n = 14,728$ frames, 30 held-out fabrics).

Band (N)	n	MAE (N)	bias (N)	within 1 N
[0, 5)	4,530	0.1971	+0.0223	0.9967
[5, 10)	3,483	0.3280	−0.0083	0.9647
[10, 15)	4,162	0.3856	−0.0741	0.9503
[15, 20)	2,339	0.4437	−0.1704	0.9235
[20, 22)	214	0.5220	−0.3943	0.8879
all	14,728	0.3252	−0.0488	0.9628

The growing negative bias at high force mirrors the thinning evidence density near the 22 N cut and motivates the 19.9 N deployment cap.

C.3 Nano17 V2: full commissioning table

Table 16 expands Section 5.6. Columns 2–4 are measured on the archived replay (frames whose reference force lies within ± 0.5 N of each target); columns 5–6 give the commissioning outcome.

Table 16 V2 commissioning outcome per plateau target.

target (N)	basis n	local bias (N)	local MAE (N)	reading (N)	error (N)
5.0	812	+0.022	0.278	5.02	0.02
7.5	682	−0.034	0.336	7.47	0.03
10.0	695	−0.035	0.374	9.96	0.04
12.5	821	−0.090	0.381	12.41	0.09
15.0	880	−0.129	0.397	14.87	0.13

A commissioning plateau averages several hundred frames, so its error is the band’s mean signed error (bias), while single-repeat scatter is bounded by the band MAE. Band-level OLS: slope 0.9882, intercept +0.077 N; plateau-mean correlation > 0.99. All five plateau errors sit far inside the 1 N acceptance bound.

C.4 Demonstration detail

Across the 24 pair–property comparisons of the four-garment run: 18 order-consistent, 6 contradictory (excluded from ranking), 0 ranking cycles, margins 0.215–6.544 (median 3.529), 0 exact ties, parse rate 0.9375 on free text (unused — forced choice is the readout). The recommendation artifact records: the resolved comparison graph, per-property ranks, the query weights, the warning that Elasticity and Softness evidence is weak on the fabric-disjoint protocol, and the statement that $n = 4$ garments / 6 pairs supports no accuracy claim. Wording is produced by an offline template in the archived run; the tactile decisions are all model outputs.

C.5 Order-reversal decomposition per cell

Table 17 applies the decomposition of Appendix A to the paired test sets. Two readings matter. First, the heldout cells show that most residual error on unseen fabrics is *position bias* rather than reverse-consistent error — on heldout-adjacent, 83 of 400 pairs pick a row position rather than a fabric, against 71 reverse-consistent pairs. Second, committed-pair accuracy exceeds row accuracy in every cell (e.g. 0.9345 vs 0.8650 on heldout gap-2): when the model commits to a physical fabric at all, it is usually right, and the aggregate is dragged down by non-commitment. Elasticity is again the outlier (committed accuracy 0.6667 on heldout gap-2, with 23 of 50 pairs position-biased).

Table 17 Order-reversal decomposition for the main paired cells (unordered pairs; each observed in both orders).
 $2/2$ = rank-correct, $0/2$ = reverse-consistent, $1/2$ = position-biased; committed accuracy = $2/2 \div (2/2 + 0/2)$.

Cell	Scope	pairs	2/2	0/2	1/2	committed acc
baseline / adjacent	all	400	309	38	53	0.8905
combined / adjacent	all	400	386	6	8	0.9847
combined / heldout adj.	all	400	391	3	6	0.9924
heldout / heldout adj.	all	400	246	71	83	0.7760
	Elasticity	100	34	26	40	0.5667
	Softness	100	52	25	23	0.6753
	Texture	100	78	9	13	0.8966
	Thickness	100	82	11	7	0.8817
heldout / heldout gap-2	all	200	157	11	32	0.9345
	Elasticity	50	18	9	23	0.6667
	Softness	50	41	2	7	0.9535
	Texture	50	48	0	2	1.0000
	Thickness	50	50	0	0	1.0000

C.6 Generation shape under the supervision ablation

Beyond the parse counts of Chapter 5 (2 unparseable generations for the verdict-restating run against 17 for the clean run, 12 of the 17 truncated at the 512-token budget), the ablation leaves a second signature: the adjacent-only run’s 400 official-test generations collapse to a single text skeleton once fabric numbers and property names are masked, with lengths spanning only 198–204 characters, while its forced and free accuracies are both 1.0000 in that cell (ceiling 0.9525). Together these show that supervision style governs the *form* of the output — ending habits, template variety — separably from decision quality, which is why this report scores decisions with the parser-free forced-choice readout.

C.7 Confidence vocabulary

Exact

combinatorial or byte-level facts independent of model stochasticity (prompt identity, look-up coverage, MD5 checks).

Measured

quantities conditional on a dataset, instrument and seed (accuracies, MAE, AUC, margins).

Interpretation

mechanistic readings consistent with, but not isolated by, the measurements (e.g. why Elasticity transfers worst).

Appendix D: Demonstration Record and Operational Notes

D.1 Full pairwise record

Table 18 lists all 24 pair–property comparisons of the four-garment run (6 unordered pairs $\times$ 4 properties, each evaluated in both presentation orders). A pair is *consistent* when both orders select the same physical garment; only consistent pairs enter the ranking of Chapter 5. Margins are fp32 forced-choice log-probability differences. No property ground truth exists for these garments, so no row of this table is an accuracy measurement.

Table 18 All 24 demonstration comparisons (F1–F4; fwd/rev = winner under each presentation order).

Pair	Property	fwd	rev	consistent	margin fwd	margin rev
F1 vs F2	Elasticity	F2	F2	✓	3.148	4.856
F1 vs F2	Softness	F2	F2	✓	5.061	2.650
F1 vs F2	Texture	F2	F2	✓	0.530	2.610
F1 vs F2	Thickness	F2	F2	✓	0.851	5.685
F1 vs F3	Elasticity	F3	F3	✓	1.612	4.109
F1 vs F3	Softness	F1	F1	✓	4.731	0.482
F1 vs F3	Texture	F1	F3	—	1.564	2.401
F1 vs F3	Thickness	F3	F3	✓	0.400	5.135
F1 vs F4	Elasticity	F4	F1	—	4.245	5.020
F1 vs F4	Softness	F4	F1	—	3.030	1.079
F1 vs F4	Texture	F4	F1	—	1.998	2.377
F1 vs F4	Thickness	F1	F1	✓	2.810	0.759
F2 vs F3	Elasticity	F2	F2	✓	1.410	3.881
F2 vs F3	Softness	F2	F2	✓	5.237	4.175
F2 vs F3	Texture	F3	F2	—	0.215	3.580
F2 vs F3	Thickness	F2	F3	—	3.478	2.024
F2 vs F4	Elasticity	F2	F2	✓	4.943	5.563
F2 vs F4	Softness	F2	F2	✓	3.997	5.254
F2 vs F4	Texture	F2	F2	✓	4.082	4.140
F2 vs F4	Thickness	F2	F2	✓	6.544	2.457
F3 vs F4	Elasticity	F3	F3	✓	5.865	5.133
F3 vs F4	Softness	F4	F4	✓	2.965	3.020
F3 vs F4	Texture	F3	F3	✓	5.851	4.402
F3 vs F4	Thickness	F3	F3	✓	6.259	1.154

Summary: 18 of 24 consistent; the six contradictions cluster on the F1–F4 pair (3 of 4 properties) and produce no ranking cycles after exclusion. Margins span 0.215–6.544 (median 3.529) with zero exact ties, confirming fp32 scoring behaved on robot-captured images exactly as on the benchmark.

D.2 Recommendation artifact

For the query “suitable for summer clothing” the recommendation layer combined the per-property ranks with weights favouring low Thickness (weight -1.0 on the Thickness rank) and high Softness

(+0.6), Texture (+0.3) and Elasticity (+0.1), selecting garment F4. The stored artifact records, alongside the choice: the six unresolved pairs excluded from ranking; the warning that Elasticity (fabric-disjoint accuracy 0.54–0.59) and Softness (0.64–0.89) evidence is weak, so a recommendation leaning on them is weakly supported; the note that $n = 4$ garments / 6 pairs supports no accuracy claim; and the provenance hashes of every input. The sentence shown to the user was produced by an offline template; every tactile decision in the graph is a project-model output.

D.3 Operational and safety notes

Three properties of the teach-and-replay controller shape safe operation and are recorded here because they are easy to get wrong:

1. **Teaching cannot be braked by software.** During demonstration teaching the position stiffness is zero — the arm follows the operator’s hand. The software stop conditions (workspace limit, timeout, watchdog, camera loss) terminate *recording*, not *motion*; the only physical protections are the operator’s grip and the real-time host’s built-in collision reaction. The taught stroke’s depth limit is the operator’s judgement; the 18–22 N peak-force check rejects a bad teach after the fact rather than preventing it.
2. **Replay can be braked.** The replayed trajectory is a policy owned by the local process; a hard stop terminates it within approximately one 20 ms control cycle. The first replay of any newly taught trajectory is nevertheless supervised with a hand on the emergency stop, because it is a trajectory that has never been executed autonomously before.
3. **Instrumentation is part of safety.** A static audit of the capture stack found and fixed three silent defects before hardware use: stale queued frames written as press frames (up to 548 ms old), capture continuing after control-loop exit, and a camera-stall check that was skipped entirely when no frame had ever arrived. After the fixes, the maximum frame–pose association error fell from 548 ms to 3.0 ms, and the run record reports median/p95/maximum pairing error rather than a single median that can hide an outlier.

These notes, together with the gates of Appendix A and the hash ledger of Appendix B, constitute the full operational record of the demonstration.